

Beliefs about the Career Costs of Children and Family Formation *

Bongseop Kim[†] Tammy Sunju Lee[‡]

December 12, 2025

[Click here for the latest version.](#)

Abstract

While existing literature highlights the career costs of childbearing as a potential contributor to low fertility in high-income countries, few empirical studies investigate how women form beliefs about these costs and whether such beliefs shape their fertility decisions. Using a sample of South Korean female workers aged 19 to 40 years, we conduct an online survey to examine the relationship between the individuals' beliefs about the career costs of children and their family formation plans. We also execute an information provision experiment to assess how objective information about the likelihood of women working post-childbirth leads to changes in beliefs and family formation plans. We find that female respondents hold pessimistic expectations about their own post-childbirth employment, partly because they underestimate the extent to which mothers in Korea remain in the workforce. These pessimistic beliefs are strongly associated with lower fertility and marriage intentions, particularly among women with strong current career prospects and higher incomes. Information provision leads to only modest changes in intentions, as many respondents interpret higher maternal employment as reflecting financial necessity rather than improved work–family compatibility. Overall, the study underscores the importance of beliefs—and their interpretation—in shaping family formation decisions.

*We thank Eleanor Jawon Choi, Syngjoo Choi, Christine Exley, Jisoo Hwang, Hoyong Jung, Chulhee Lee, Jungmin Lee, Tanya Rosenblatt, and Basit Zafar for their valuable comments and feedback. We are also grateful to participants at ESWC 2025, PAA 2025, and seminar audiences at the University of Michigan and Seoul National University. We additionally appreciate helpful discussions with Daniele Vignoli. This study was supported by the National Research Foundation of Korea, the Korea Experimental and Behavioral Economics Seed Grant, the Social Science Student Research Grant through Seoul National University, and the PSC Small Grant award through Institute for Social Research at University of Michigan. We received the IRB approval from Seoul National University (IRB No.2401/001-007). This study is registered at the AEA Registry (AEARCTR-0013738).

[†]Department of Economics, Seoul National University. Email: bongseop@snu.ac.kr

[‡]Department of Economics, University of Michigan - Ann Arbor. Email: tammlee@umich.edu

1 Introduction

Across high-income countries, birth rates have declined to historic lows. Among those, South Korea stands out, with its total fertility rate dropping to a record low of 0.72 in 2023. This unprecedented decline calls for new explanations of fertility choice. Recent research has highlighted the compatibility of family and career as a leading explanation (see [Doepke et al. \(2023\)](#) for a review). As women increasingly seek to combine career ambitions with family life ([Goldin, 2021](#)), the ability to reconcile work and family responsibilities has become central to fertility choice.¹ This perspective is echoed in survey evidence: in a recent Korean government survey, respondents most frequently cited ‘difficulty balancing work and family’ as the main reason for the country’s record-low fertility rate. Reflecting this, the government has made work–family compatibility a top priority, allocating more than 80% of its expanded 2024 low-birth-rate budget to this goal ([Presidential Committee on Ageing Society and Population Policy, 2023, 2024](#)).

Despite growing attention to the role of career costs in fertility decisions, few studies examine how individuals form *expectations* about these costs. This question is particularly salient in Korea, where fertility rates are exceptionally low and mothers have historically faced severe work–family tradeoffs ([OECD, 2024a](#)). Yet conditions have changed rapidly: as shown in [Figure 1](#), the labor force participation rate of mothers with children aged 0–6 has risen sharply only since around 2010. In this context, recent cohorts may continue to base their expectations on past conditions, leading them to systematically *overestimate* the career costs of motherhood—even as opportunities to remain in the workforce have expanded. As [Goldin \(2025\)](#) highlights, macroeconomic change can outpace shifts in beliefs, values, and traditions.²

¹Figure [A1](#) illustrates a negative cross-country correlation between the child penalty and the average total fertility rate in high-income countries. In countries with a minor child penalty, where work-life compatibility for women is more achievable, fertility rates are generally higher, whereas in countries with a significant child penalty, such as Korea, fertility rates are lower.

²Relatedly, [Kuziemko et al. \(2018\)](#) show that younger cohorts of women in the US and UK underestimate the employment costs of motherhood because they form expectations from their parents’ generation, when balancing work and family was relatively easier.

In this paper, we address four questions: (1) How do women of childbearing age perceive the career costs of having children? (2) To what extent are these expectations linked to their family formation plans? (3) Are these expectations accurate reflections of current societal conditions? (4) If not, does providing information alter women’s plans for family formation?

To examine these questions, we conduct an online survey of about 1,900 Korean female workers aged 19 to 40. We begin by eliciting respondents’ perceptions about the career impact of motherhood on Korean women. Specifically, we ask respondents to estimate the percentage of women in Korea who are working a few years after having their first child. We then ask respondents about their expectations regarding the career implications of motherhood, followed by questions on fertility intentions and the perceived likelihood of having children within three, five, and ten years, as well as analogous questions on marriage.

Respondents are then randomly assigned to one of two treatment groups or to a control group. One-third receive accurate information on the share of Korean women who are working two years after childbirth, based on data from Statistics Korea (2-year treatment group). Another third receive similar information for five years post-childbirth (5-year treatment group). The remaining third receive information unrelated to parenthood: the percentage of Korean female workers with a college degree (active control group). After the information treatments, we re-elicite respondents’ beliefs about the career costs of childbearing, as well as their family formation intentions and expectations.

Our key findings are as follows. First, we find that female respondents substantially overestimate the career costs of motherhood. While 69% and 77% of Korean women are employed two and five years after childbirth, respectively, respondents estimate these figures to be just 36% and 39% on average. Over 90% of respondents underestimate the actual proportion of women who remain employed post-childbirth. This pattern persists across industries, even when questions are tailored to respondents’ own sectors. This pattern is not due to disutility from work because more than 90% women agreed that it is more ideal for mothers to work post child birth.

We also document suggestive evidence that personal exposure to family-compatible environments is associated with more accurate beliefs. In particular, respondents who grew up with a working mother or who currently work at firms with generous parental leave policies report estimates closer to the population average. These two factors explain a substantial share of the variation in individual beliefs about women’s career costs after childbirth. This suggests that direct or observed experience with supportive work-family arrangements may play a key role in shaping beliefs.

Second, we find that inaccurate beliefs about the career costs of motherhood contribute to pessimistic expectations about one’s own future employment. Respondents who underestimate the share of Korean women employed after childbirth are less likely to expect that they themselves would remain in the workforce after having a child. Specifically, a 1 percentage point decrease in beliefs about mothers’ post-childbirth employment is linked to a 0.5 percentage point decrease in respondents’ own expected probability of working after having a child.

Third, these pessimistic expectations, in turn, are strongly associated with lower intentions regarding fertility and marriage. A shift from full confidence in continued employment after childbirth (100%) to no expectation of doing so (0%) is associated with a 23% decline in overall fertility intentions relative to the mean. It also corresponds to 7, 14, and 17 percentage point decreases in the probability of having children within 3, 5, and 10 years, respectively—equivalent to relative declines of 23%, 34%, and 34%. These effects are most pronounced among women with strong career trajectories—those with higher earnings and positive career outlooks—suggesting that expectations of career interruption act as a substantial deterrent to childbearing among women facing higher opportunity costs.

Finally, we show that correcting inaccurate beliefs through an information provision has modest effects on fertility plans. On average, treated respondents increase their expected probability of having a child by 1.3 points within three years and 1.2 points within ten years, relative to control means of 30.6% and 52.7%. Instrumental variable estimates suggest that a

10 percentage point increase in perceived post-birth employment rates leads to approximately a 0.7 and 0.6 percentage point increase in the probability of having a child within 3 and 10 years. We find no significant effects on 10-point fertility intention measures.

Follow-up questions reveal that the modest effect stems from respondents’ reluctance to update their priors about work–family compatibility. Many explained their low estimates of mothers’ post-birth employment as a reflection of persistent incompatibility between work and family. When provided with higher-than-expected employment rates, however, they tended to interpret these as evidence of financial necessity rather than improved compatibility. This interpretation limits the extent to which new information translates into changes in fertility and marriage intentions, as people do not view it as evidence of improved work–family compatibility. It suggests that information alone may be insufficient to shift behavior without credible narratives about structural change.

Taken together, our findings suggest that outdated or pessimistic beliefs about career costs can act as a barrier to fertility even in contexts where structural conditions are improving. Enhancing work–family compatibility may not fully translate into increased fertility unless individuals also perceive—and internalize—these changes. In rapidly evolving policy environments like Korea’s, shaping accurate and contextually meaningful beliefs may be a critical complement to structural reforms aimed at addressing low fertility.

We contribute to the economics of fertility literature by documenting women’s *beliefs* about work–family compatibility and examining how these beliefs relate to fertility intentions. Recent studies have shown that children impose significantly greater setbacks for women than men, commonly referred to as the “Child Penalty” or “Motherhood Penalty” (see the review in [Cortes and Pan \(2023\)](#)). Numerous studies have extended the fertility decision model by incorporating these career-family tradeoffs women face (see the review in [Doepke et al. \(2023\)](#)). [Feyrer et al. \(2008\)](#) indicates that in countries at an intermediate stage - where women have improved labor market opportunities but their household status lags - the career costs of women having children become particularly high, potentially lead-

ing to lower fertility rates. [Doepke and Kindermann \(2019\)](#) develop a household bargaining model where the distribution of the burden of child care between mothers and fathers plays a crucial role in determining fertility. Other studies highlight the role of labor market institutions in influencing the career costs of childbearing for women and explore the relationship between these institutions and fertility ([Adserà, 2004](#); [Rocha and Fuster, 2006](#); [Lopes, 2024](#); [Guner et al., 2024](#)). Family policies, such as public childcare, are also considered influential in mitigating the career penalty associated with motherhood, thus impacting fertility rates ([d’Albis et al., 2017](#); [Olivetti and Petrongolo, 2017](#)). While prior studies implicitly assume that women incorporate the expected career costs of childbearing into their fertility decisions, our study explicitly examines these beliefs.

A few recent studies have examined individuals’ beliefs about post-childbirth labor supply and their implications. [Kuziemko et al. \(2018\)](#) show that women in the US and UK tend to underestimate the employment costs of motherhood, leading to overly optimistic schooling choices. [Wiswall and Zafar \(2021\)](#) show that female students’ beliefs about their post-childbirth labor supply influence their choice of major. [Gong et al. \(2022\)](#) find that men and women hold different expectations about their labor supply and family outcomes, and that these expectations closely match realized outcomes when surveyed ten years later in the US. While these studies focus on how beliefs are formed and how they shape education and labor market choices in the US and UK, none examine their relationship to fertility intentions. Our paper extends this literature by documenting women’s beliefs about work–family compatibility in Korea—which until recently exhibited substantial child penalties—and by analyzing how these beliefs relate to fertility intentions.

Our paper also contributes to the literature that attempts to identify new factors explaining the low fertility rates in high-income countries. These factors include education fever ([Kim et al., 2024](#)), housing prices ([Yi and Zhang, 2010](#); [Dettling and Kearney, 2014](#)), traditional social norms ([Myong et al., 2021](#)), peer effects ([Rossi and Xiao, 2022](#)), and shifting priorities ([Kearney et al., 2022](#)). To the best of our knowledge, we are the first to propose that

people’s misperceptions may be one of the reasons for low fertility in high-income countries.

Finally, our paper relates to labor studies that utilize survey methodology to investigate how individual beliefs influence labor-related outcomes. (see the review in [Stantcheva \(2023\)](#), as well as [Haaland et al. \(2023\)](#) and [Fuster and Zafar \(2023\)](#)). These topics include school choice ([Bleemer and Zafar, 2018](#)), major choice ([Wiswall and Zafar, 2015, 2021](#)), labor market participation ([Boneva et al., 2023](#); [Costa-Ramón et al., 2024](#)), job search ([Jäger et al., 2024](#)), and parental investments in the child’s human capital ([Deshpande and Dizon-Ross, 2024](#)).

The rest of the paper is organized as follows. Section 2 outlines the conceptual framework motivating our survey design and analysis. In Section 3, we detail data collection and survey design. Section 4 examines people’s beliefs about the career costs of children and in Section 5, we explore family formation plans and their correlation with beliefs. In Section 6, we analyze information provision experiments. Section 7 concludes.

2 Conceptual Frameworks

How do individuals’ beliefs about the current compatibility of work and family life shape their fertility plans? To guide our empirical analysis, we develop a simple conceptual framework that connects beliefs about career costs of childbearing to family formation decisions. This framework helps organize our survey design and interpretation of results, as illustrated in Figure 2.³

Fertility decisions are inherently forward-looking: individuals weigh current circumstances against expectations about the future environment for child rearing and the anticipated costs associated with having children. Recent models emphasize this forward-looking nature, where fertility choices depend on expected utility that incorporates both household preferences and future constraints ([Doepke and Kindermann, 2019](#)). One critical component of these expected costs—especially salient for women—is the potential disruption to employment after having a child. As highlighted in the literature, having children often leads to interruptions in

³A formal model is provided in Appendix 2.

women’s careers, reduced earnings, and slower career progression (Kleven et al., 2019, 2023). In contexts where work and family goals are less compatible, these anticipated costs are particularly salient. Thus, women’s beliefs about the likelihood of maintaining employment after childbirth—or about the overall compatibility of work and family—would play a central role in shaping their fertility plans (see descriptive evidence in Doepke et al., 2023).

To formalize the role of expected costs, we distinguish between two types of beliefs: *population beliefs* and *self beliefs*. Population beliefs refer to individuals’ perceptions about the average experience of women in their society—for example, the share of Korean women who remain employed after childbirth. These beliefs serve as a mental model or summary of the broader social environment and policy landscape. Drawing on Bleemer and Zafar (2018), we conceptualize population beliefs as the informational foundation from which self beliefs—individuals’ expectations about their own career prospects—are derived. Self beliefs directly inform personal decision-making, while population beliefs shape those self beliefs. Population beliefs, in turn, are influenced by a range of factors including socio-demographic characteristics, current job and workplace environments, and formative experiences such as growing up with a working mother.⁴

Understanding how individuals form population beliefs is important because these beliefs are often misaligned with objective realities (Kuziemko et al., 2015; Alesina et al., 2018, 2023). If people overestimate the career penalties of childbearing for women in general, they may develop unduly pessimistic expectations about their own prospects—even in environments where policies or norms have shifted. Such misperceptions can suppress fertility intentions when individuals fail to update their beliefs in line with improved structural conditions.⁵

In the following section, we first examine how Korean women perceive these average career

⁴See, for instance, Kuziemko et al. (2018) for evidence that early-life experiences can shape individuals’ beliefs of work–family compatibility.

⁵Bursztyn et al. (2020) provide related evidence in a different context: when husbands’ underestimated support for female work was corrected, wives’ employment probability increased, underscoring how misperceptions can directly constrain behavior.

costs and present evidence on the relationship between population beliefs, self beliefs, and fertility plans. We then provide experimental evidence on how information treatment shapes beliefs and fertility plans, allowing us to identify the causal link. While women’s selection into jobs may reflect underlying fertility preferences (Adda et al., 2017), our focus is on how correcting perceived realities affects fertility plans conditional on these pre-existing choices. We account for this by holding constant current job and workplace characteristics, which were determined prior to the information intervention. Our aim is to isolate how differences in beliefs—rather than in structural constraints—drive variation in family formation plans among otherwise similar individuals.

3 Survey

3.1 Data Collection

In June 2024, we conducted an online survey of 1,500 female employees aged 19-40 who were born in South Korea, using Embrain, a leading commercial survey company in Korea.⁶ Embrain operates quota sample panels, similar to platforms such as Qualtrics and Prolific.⁷ Embrain has a panel of approximately 1.8 million Koreans who have agreed to provide their personal information. We selected individuals meeting the criteria from this panel and invited them via email to participate in the survey. Participation was voluntary, and each complete survey had a compensation of about 2,000 KRW. The full survey questionnaire is described in Appendix 4. Prior to data collection, we submitted a pre-analysis plan to the AEA RCT Registry: <https://www.socialscienceregistry.org/trials/13738>.

⁶A parallel survey of 1,500 men was also conducted; those results are reported in a separate paper.

⁷Quota sampling panels are a form of non-probability sampling, where advertisements are sent to individuals who meet specified criteria and the survey continues until quotas are filled. Compared to probability sampling, quota sampling is less costly and is widely used in practice (Stantcheva, 2023). However, quota sampling carries risks regarding representativeness, which we discuss more in Section 3.4.

3.2 Survey Design

Individual, Job, and Workplace Characteristics: We collect information about respondents’ characteristics, including gender, age, employment status, income, marital status, number of children, highest level of education achieved, place of residence, and political leaning. We also gather detailed information on job characteristics such as occupation, industry, firm size, department size, and working hours. Additionally, we inquire about the extent to which parental leave benefits, working from home, and flexible working arrangements are utilized at their current workplaces. For respondents with spouses, we also collect data regarding the spouse’s employment status, occupation, industry, income, and highest level of education achieved.

Beliefs Elicitation: Respondents are asked about their perception of the probability of female workers in Korea continuing to work after having their first child. The question is as follows: *“What do you think is the probability that a female worker in Korea, who became pregnant with her first child in 2015, would be working [2 years/5 years] after giving birth?”* This question was posed twice, first with a two-year horizon and then with a five-year horizon following the child’s birth. We refer to these as *population beliefs* since they reflect individual beliefs about the average effect of childbearing on mothers’ employment across the Korean population. We further tailor population beliefs to specific industries. Respondents are asked about the probability of employees in their respective industries to work after having their first child. We focus on two- and five-year horizons. The two-year mark captures short-run employment responses, while the five-year horizon roughly coincides with the period when children approach school age, a stage at which many Korean mothers experience career interruptions (Choi et al., 2022). Although children typically enter elementary school at age six, we use the five-year interval because our administrative employment data only extend through 2021, making it the latest horizon we can consistently validate. To measure the uncertainty in their population beliefs, respondents also rate their confidence on a 5-point Likert scale.

We then ask about respondents’ expectations regarding the career costs they and their spouses may face after having children. They are asked about the expected probability that they could continue to work after having children. The question is as follows: *Imagine a scenario where you have your first child (or an additional child if you already have one). What do you think is the probability that you will continue to work for [2 years/5 years] after the child is born?* These are referred to as *self beliefs* since they relate to the respondents’ beliefs on their personal career costs associated with childrearing. To capture different dimensions of career impact, we also ask about the expected probability of remaining at the current workplace, expected wage levels and average working hours, and the probability of re-employment after leaving a job due to childbirth.⁸

Family Formation Plans: Our main outcome of interest is respondents’ fertility plans. Respondents are asked about their intentions to have a child on a scale of 0 to 10, the number of children they wish to have, and their timeline for having children. They are also asked about their likelihood of having children within three, five, and ten years.⁹ Because childbirth in Korea predominantly follows marriage (OECD, 2024b), we also ask respondents about their marriage intentions, the timing they anticipate, and the likelihood of marrying within specific horizons.

Gender Norms and Related Beliefs: Respondents are asked about their views on the role of mother, namely whether they agree with the following statements: A husband’s job is to earn money, and a wife’s job is to take care of the home and family; When a mother works, her preschool-aged children suffer significantly; Even after getting married, it is better for a woman if she continues to work. Additionally, we ask about respondents’ beliefs that are related to their expected career costs after having children. These include expectations of spousal support in childcare and potential parental assistance with childcare. Finally,

⁸These additional questions are posed only after respondents receive the information treatment described in Section 6. Consequently, only the control group is used for correlation analysis of these outcomes.

⁹Previous research indicates that stated fertility plans are highly predictive of future childbearing (Doepke and Kindermann, 2019; Duvander et al., 2020).

respondents are asked about their expected career advancement prospects if they continue working.

Experiences: Respondents are asked about their mother’s employment status during their childhood and the education levels of both parents. We also collect data on whether they have siblings who have given birth, including the child’s birth year and the sibling’s current employment status. This inquiry extends to both brothers and sisters. Additionally, respondents are asked about the number of female colleagues in their department who have recently given birth and how many of them remain employed at the company afterward.

3.3 Data Quality

We implement several measures to ensure the quality of responses. In the survey introduction, we clearly state our institutional affiliation and stress the importance of honest responses for the success of the research. We further emphasize that truthful responses contribute to the broader public good. For questions on population beliefs, where correct answers can be verified, we incentivize respondents to make careful and accurate guesses by offering a 3,000 KRW reward for answers within 1% of the actual statistic. We monitor the time taken by respondents to complete the survey and exclude those who complete the survey either too quickly (under five minutes) or too slowly (over one hour), representing approximately 3.7% of the sample. Additionally, we exclude respondents who consistently select the extremes (0 or 100) in belief elicitation questions, considering those respondents being inattentive (accounting for 1.7% of respondents).

3.4 Sample Characteristics

Our final sample consists of 1,419 respondents. Table 1 displays the socioeconomic characteristics of the sample: 23% are married; 10% have children; 36% reside in Seoul, the capital of Korea; and 73% have a university degree. The average age is 31 years, and the average

monthly wage stands at 3.04 million KRW. The proportion of respondents working in firms with over 100 employees is 35%, and 39% report having access to parental leave in their workplace. A majority of respondents (58%) report having a positive outlook on their career prospects if they continued working. Furthermore, about 90% of respondents state that they believe it is beneficial for women to remain in the workforce after marriage, indicating widespread support among women for continued employment.

Table A1 compares the summary statistics from our survey to those from a nationally representative sample of South Korea employees based on Korean Working Condition Survey (KWCS).¹⁰ Our sample is representative in terms of gender and age. However, individuals with higher education, urban residence, and higher incomes are disproportionately represented—a pattern commonly observed in quota sampling. In addition, employees working at large-sized firms are more prevalent in our sample. Conservatively, one might consider our results as more representative of high-earning, more educated, urban populations. Given that prior research has documented that career-related concerns are especially pronounced among women in these demographic groups, focusing on them is both natural and policy-relevant (Goldin, 2021). Nevertheless, to address concerns regarding external validity, we follow previous studies and apply reweighting methods to align our sample with the population distributions (Stantcheva, 2023).¹¹ As a robustness check, we create weights to match our sample to the KWCS in Table A1, and produce reweighted results. Details of the weighting procedure and the results are provided in Appendix 3. Reweighting has no appreciable impact on the main results; therefore, we focus on the unweighted results throughout the paper.

¹⁰The KWCS is conducted every three years by the Occupational Safety and Health Research Institute (OSHRI) on a representative sample of approximately 50,000 individual workers. Its questionnaire is based on the European Working Conditions Survey conducted by Eurofound.

¹¹Grewenig et al. (2023) shows that re-weighted non-probabilistic online surveys can produce response patterns that are mostly indistinguishable from those of offline surveys. They emphasize that re-weighting on-liners can be a pragmatic, economical solution for applied researchers in many contexts to depict preferences or beliefs of the entire population.

4 Beliefs about Career Costs of Children

In this section, we present descriptive evidence on respondent beliefs about the impact of having children on mother’s employment and the correlation of these beliefs with other variables.

4.1 Population Beliefs

Figure 3 shows the distribution of population beliefs regarding the proportion of Korean women who remain employed after childbirth. Panel (a) shows beliefs about women working two years after first childbirth, and panel (b) shows beliefs about women working five years after first childbirth. Respondents’ beliefs are extremely right-skewed: while the actual proportion of female workers who continue working two years after childbirth is 69% in Korea, respondents believe it to be only 36% on average. For five years, the actual proportion is 77%, whereas respondents believe it to be 39%.¹² Approximately 90% of respondents report beliefs below the actual share of women working two years after childbirth, and 92% do so for the five-year mark, indicating widespread misperceptions about mothers’ labor force participation in Korea. These results are not due to inattentive survey responses, as incentives are offered for accurate responses. Furthermore, we do not observe a comparable right-skewed deviation when respondents are asked about the proportion of Korean male workers continuing work post-childbirth (Figure A2).

These misperceptions persist even when respondents are asked about workers within their own sectors. Tables A2 show sector-specific results, showing beliefs about the post-childbirth employment rates of women in respondents’ respective sectors. The actual share of women working two and five years after childbirth varies substantially across sectors. For example, in the Public Administration, Social Security, and Education sector, 81% of female workers are employed two years after childbirth, whereas in the Healthcare and Social Work

¹²The actual proportions are calculated using the employment registry from Statistics Korea. Individuals on parental leave are classified as not working.

Activities sector, the proportion is 65%. However, in all sectors, over 80% of respondents provide estimates below the actual share of women working after childbirth. Misperceptions are especially pronounced in Manufacturing and Construction, where more than 95% of respondents report figures below the true proportion.

Which factors, then, predict individuals’ population beliefs? Figure 4 presents regressions of the two- and five-year population beliefs on individual characteristics, job and workplace traits, and personal experiences. First, we find that personal experiences are important predictors. A mother’s employment status during a respondent’s childhood is one of the strongest predictors: respondents who reported that their mothers worked during their childhood show population beliefs about 3 percentage points higher, underscoring the enduring impact of early-life experiences on belief formation. Additionally, having coworkers with children in the past five years is associated with beliefs about 3 percentage points lower.

Age is another significant factor: respondents in the 30–40 age group report beliefs about 2.5 percentage points lower than those aged 19–29. Education shows a similar pattern, with college graduates reporting estimates roughly 4 percentage points lower than those without a college degree. Finally, respondents with more progressive gender norms also report estimates about 4 percentage points lower.

In terms of job and workplace characteristics, being a permanent employee is associated with population beliefs about 3 percentage points higher than those of temporary workers. Similarly, access to parental leave at one’s current workplace is linked to beliefs roughly 4 percentage points higher. These patterns suggest that workplace conditions are closely related to the formation of population beliefs.

To further assess the relative importance of each predictor, we conduct a Shapley decomposition of the variance in population beliefs (Table 2).¹³ The results highlight that early-life experiences and current workplace environments explain the largest share of variation in

¹³Shapley decomposition is a game-theoretic method that attributes the explained variance of an outcome to different sets of predictors in proportion to their marginal contributions across all possible orderings of inclusion. It allows assessing the relative importance of factors in explaining variation in outcome.

beliefs. Having a working mother during one’s childhood accounts for about 25 percent of the variation, while access to parental leave in the workplace contributes another 25 percent. By contrast, demographic characteristics such as marital status, age, and number of children explain only a small fraction of the variance, each accounting for less than 5 percent. These findings underscore that both formative experiences and institutional features of the workplace are the dominant drivers of misperceptions regarding women’s post-childbirth employment, rather than demographic correlates alone.

4.2 Self Beliefs

We next examine respondents’ beliefs about their own career costs associated with having children, focusing on the expected probability of remaining in the workforce. Panel A of Table 3 reports summary statistics of reported beliefs. Respondents, on average, believe they have a 59% and 65% chance of continuing to work for two and five years after childbirth, respectively. Although estimates are higher than respondents’ population beliefs, they remain roughly 10 percentage points lower than the actual post-childbirth employment rates of women in Korea. Furthermore, 47% and 38% of respondents estimate their probability of remaining employed is below fifty percent at two and five years post-childbirth, respectively. This reflects a widespread pessimism about their career prospects following childbirth.¹⁴

This pessimism among women is even more marked in their responses about the probability of remaining at their current workplace. On average, respondents perceive a 50% and 45% chance of maintaining their position at their current workplace at two and five years post-childbirth, respectively, with 61% and 65% believing their chances are below fifty percent. Respondents also estimate only a 54% chance of re-entering the workforce should they need to leave their job. Overall, respondents expect to face substantial career costs after childbirth.

¹⁴In the Berea Panel Study (BPS) conducted in the United States, when asked about their expected probability of working at age 28, female respondents, on average, report a 79% probability of working when their children are aged 0-2 and an 83% probability when their children are aged 3-5. For male respondents, the probability is 96% in both cases (Gong et al., 2022).

Next, we examine how individuals’ beliefs about average Korean women (*population beliefs*) relate to their own expected employment prospects after childbearing. We refer to the expected probability of continuing to work after childbirth as *self-beliefs*, which correspond directly to the population beliefs measure. Figure 5 displays a binscatter plot of respondents’ self beliefs against population beliefs, illustrating a clear positive relationship. Regression analyses show that female respondents’ self-beliefs are closely linked to their population beliefs, even after controlling for individual characteristics, job and workplace traits, personal experiences, and other beliefs such as expected familial support for childcare. At both the two- and five-year horizons, a 1 percentage point increase in population beliefs about women working after childbirth corresponds to roughly a 0.5 percentage point increase in respondents’ own expected probability of working (Table 4). These findings suggest that previously documented misperceptions about the impact of childbearing on women’s employment may contribute to pessimistic expectations about one’s own prospects after childbirth. When we re-estimate the regression using sector-specific population beliefs instead of overall beliefs, the coefficients remain similar, ranging from 0.4 to 0.5 (Table A4).

Table A3 presents the regression coefficients for other related variables. Women with a college degree tend to have self beliefs that are 3 to 4 percentage points higher, and those earning above the average income have beliefs that are 3 to 5 percentage points higher. Having a working mother is linked to a 3 percentage point increase in self beliefs. Women adhering to more progressive gender norms exhibit a 6 to 7 percentage point increase in self beliefs, likely indicating a stronger commitment to remain in the workforce. Additionally, the ability to freely take parental leave at one’s current workplace emerges as one of the strongest predictors of self-beliefs: Women who can freely take parental leave form beliefs that are approximately 6 percentage points higher, suggesting that the workplace environment is crucial in shaping their expectations. Also, women who report a lack of support from their spouse or parents in childcare tend to have self-beliefs that are 3 to 4 percentage points lower, emphasizing that the allocation of childcare responsibilities significantly influences women’s

expectations regarding career interruptions post-childbirth.

5 Family Formation Plans

This section explores the link between respondents’ self beliefs and their family formation plans. Table 5 presents summary statistics on family formation plans. On a scale from 0-10 for fertility intention, respondents report an average of 4.6. Regarding the likelihood of having children within the next three, five, and ten years among childless respondents, respondents report probabilities of 28%, 40%, and 50%, respectively. They on average desire 1.2 children. On a scale measuring marriage intentions from 0-10, respondents report an average of 5.9. When asked about probabilities of marrying within the next three, five, and ten years, they report chances of 43%, 55%, and 65%, respectively.

5.1 The Link Between Beliefs and Family Formation Plans

To explore the relationships between beliefs and family formation plans, we regress respondents’ family formation plans on their self beliefs using the following model:

$$Y_i = \beta_0 + \beta_1 S_i + \gamma X_i + \epsilon_i \quad (1)$$

where Y_i represents fertility and marriage plans for individual i , S_i represents a self beliefs, i.e., the expected probability of being able to continue working, and X_i denotes controls for individual, job, and workplace characteristics. This allows us to capture the residual effect of expectations about own career costs post-childbirth on family formation plans, conditional on individual, job, and workplace-related characteristics.¹⁵

Table 6 presents the estimates of β_1 for each outcome. Table 6(a) shows that self beliefs are significantly and positively correlated with both fertility intentions and the expected probability of having children. Shifting from no expectation of continuing to work two years

¹⁵The results remain robust after controlling for respondents’ spouse beliefs.

post-childbirth (0) to fully expecting to continue (100) correlates with a 0.7 point increase in fertility intention (a 15% increase from the mean), and increases in the expected probability of having children within 3, 5, and 10 years by 7.8 percentage points (28% increase), 11.6 percentage points (29% increase), and 12.0 percentage points (24% increase), respectively. The positive relationship is more pronounced regarding beliefs about continuing to work five years after childbirth. Shifting from no expectation of working five years post-childbirth (0) to fully expecting to continue (100) correlates with a 1.1 point increase in fertility intentions (a 24% increase). It also correlates with increases in the expected probability of having children within 3, 5, and 10 years by 6.5 percentage points (23% increase), 13.6 percentage points (34% increase), and 17.2 percentage points (34% increase), respectively. Furthermore, a five year projection in population beliefs corresponds to a 0.2 point rise in the desired number of children (17% increase). The larger coefficient for the expectation of working five years after childbirth, compared to two years, indicates that women place more importance on their employment situation at later stages post-childbirth when planning their families.

Table 5(b) demonstrates that marriage intentions and the expected probability of marriage also exhibit a positive correlation with self-beliefs, which intensifies when considering beliefs about continuing work 5 years after childbirth. Conversely, no significant relationships are found between self beliefs and the timing of childbirth and marriage (Table A5).

We conduct a heterogeneity analysis to identify the groups for whom beliefs play a particularly important role. Table 7 reports the results by respondents' current career prospects. We divide the sample into two groups: those who are positive about their current career prospects ("very good" or "good"), and those who are neutral or negative ("neutral," "negative," or "very negative"). Panels A and B present the results for the group with positive prospects and the group with non-positive prospects, respectively. We find that the positive correlation between self beliefs and family formation plans is more pronounced among those with positive career prospects. Moreover, when dividing the sample by monthly income into a higher income group (monthly income of 3 million KRW or above) and a lower income group

(monthly income below 3 million KRW), the positive correlation is more pronounced in the higher income group (Table 8). These results indicate that in groups facing a higher opportunity cost of career interruptions, expectations about the possibility of continuous employment play a particularly critical role in family formation decisions. When conducting regression by redefining self-beliefs based on expectations of continued employment at the same workplace and the probability of reemployment, we also find that the coefficients remain significantly positive (Tables A6 and A7).¹⁶ Overall, our results indicate that women’s expectations of career interruptions after childbirth are closely linked to their family formation plans.

6 Experimental Analysis

In this section, we present findings from the information provision experiment in which some respondents were informed of the actual rate of women working after childbirth in Korea. This variation helps evaluate whether providing accurate information can correct misperceptions and serve as a policy lever to influence fertility intentions.

6.1 Information Treatment

After being asked about beliefs and family formation plans, respondents are randomly assigned to one of three treatment arms. One-third of the respondents (2-year treatment group) receive the following information:

Based on data from Statistics Korea, among 100 Korean female workers who became pregnant with their first child in 2015, approximately X out of 100 were still employed two years post-childbirth. In the [respondent’s own (or spouse’s) sector], among 100 Korean female workers who became pregnant with their first child in 2015, approximately Y out of 100 were still employed two years post-childbirth.

Our information treatment includes several notable features. First, it comprises both gen-

¹⁶Expected wages and working hours do not exhibit a significant relationship.

eral information applicable to all workers and tailored information specific to the respondent’s own sectors. By providing sector-specific information that is directly relevant to respondents’ circumstances, we aim to increase the credibility and impact of the information. Second, our design introduces variation in the values received by respondents within the same treatment group. Although the actual employment rate of female workers who gave birth in 2015 is 69% two years later, the values received by respondents range from 56% to 85%, depending on how the sample of 100 women is selected from the population. This variability within the treatment group generates additional variation in treatment intensity, allowing us to better identify the effects of information provision.

Another one-third of the respondents receive similar information but for five years after childbirth (5-year treatment group). The remaining one-third are provided with statistics on the percentage of college graduates among Korean female workers (Control group). We employ an active control group design, allowing us to control for potential effects stemming from merely receiving information ([Haaland et al., 2023](#)).

After viewing the assigned information, respondents must enter the number that matches the provided information to continue with the survey. They can revisit the information by clicking a button on the screen if necessary. Once the percentages are correctly entered, all respondents answer the questions on belief elicitation and family formation plans again.

Table [A8](#) presents socioeconomic statistics for each group. Columns (1)-(3) show the mean for the treatment group and control groups respectively, indicating no systematic differences between these groups.

6.2 Main Empirical Specification

To examine the effect of information on beliefs and family formation plans, we estimate the following specifications:

$$Y_i = \beta_0 + \beta_1 T_i^2 + \beta_2 T_i^5 + \gamma X_i + \epsilon_i \quad (2)$$

Y_i denotes the outcome variable of interest, which includes belief and family formation plans measured after receiving the information.; T_i^2 is a dummy that takes value one if individual i belongs to the 2-year treatment group; T_i^5 is a dummy that takes value one if individual i belongs to the 5-year treatment group; X_i is a set of control variables, which include individual, job, and workplace characteristics.

6.3 Treatment Effects on Beliefs

Table 9 presents regression results for post-treatment beliefs about the impact of children on employment. Columns (1) and (2) of Panel A show that the treatment significantly increases population beliefs, i.e., beliefs about the proportion of Korean female workers who are employed after childbirth. For respondents in the 2-year treatment group, population beliefs for two and five years increase by approximately 19 percentage points (47% increase) and 20 percentage points (45% increase), respectively. In the 5-year treatment group, these beliefs increase by approximately 20 percentage points (49% increase) and 21 percentage points (49% increase) for two and five years, respectively. These results demonstrate that respondents update their beliefs not only for the specific period covered by the information but also across different time spans. It is important to note that while participants do revise their beliefs, they do not adjust to the full extent of the information provided. The average discrepancy between participants' initial population beliefs and the information received is 33% for two-year population beliefs and 38% for five-year population beliefs. Both the 2-year and 5-year treatment groups, on average, revise their beliefs by approximately half of this gap.

Columns (3) and (4) show that treatment significantly increases respondents' self beliefs, i.e., the expected probability of continuing to work after childbirth, but the effect size is modest. For respondents in the 2-year treatment group, two- and five-year self-beliefs increase by approximately 4 percentage points (a 6% increase) and 3 percentage points (a 5% increase), respectively. In the 5-year treatment groups, two- and five-year self-beliefs each increase by

approximately 5 percentage points (an 8% increase) and 4 percentage points (a 7% increase). To better interpret the causal effect sizes, we employ a 2SLS framework that instruments posterior population beliefs with the random treatment assignment.¹⁷ We find that a 1 percentage point increase in both two-year and five-year population beliefs leads to about 0.2 percentage points increase in both two-year and five-year self beliefs (Panel B).

6.4 Treatment Effects on Family Formation Plans

Next, we examine whether the information treatment leads to changes in respondents' family formation plans. Table 10 presents regression results for respondents' post-treatment family formation plans. Table 10(a) shows that treatment has modest and somewhat inconsistent effect on fertility plans. On average, treated respondents increase their expected probability of having a child by about 1 to 2 percentage points. Instrumental variable estimates suggest that a 10 percentage point increase in perceived post-birth employment rates leads to a 0.6 to 0.7 percentage point increase in the probability of having a child. However, we find no significant effects on fertility intentions or the desired number of children. We also find no statistically significant differences in marriage plans between the treatment and control groups (Table 10(b)).¹⁸

6.5 Why Are Treatment Effects Not Significant?

While beliefs are strongly correlated with family formation plans, inducing positive updates in self-beliefs does not generate corresponding shifts in those plans. We conduct additional analysis to explore potential explanations for this discrepancy and find that respondents' interpretations of the information are a likely driver: they tend to view higher-than-

¹⁷The first-stage F -statistics correspond to 395 for two-year population beliefs and 365 for five-year population beliefs, respectively.

¹⁸Overall, changes in family formation plan responses among treatment groups are infrequent. For instance, only 7% of respondents in the treatment group increase their fertility intention post-treatment compared to pre-treatment, while the majority maintain their initial responses. In the control group, 8% of respondents increase their fertility intention

expected maternal employment as reflecting financial necessity rather than improvements in the work–family policy environment.

First, we assess whether the limited effects on family formation plans are due to a small first-stage effect—that is, whether the shift in self-beliefs is simply too small to induce further updates. To examine this, we test whether the effect is more pronounced among respondents whose first-stage impacts are likely larger: those with prior two- and five-year self-beliefs below the medians of 65 and 75, respectively (“Pessimistic”), compared to those above the median (“Non-Pessimistic”). We then replicate the analysis with an interaction term between the treatment and the pessimism indicator. As expected, respondents with lower prior self-beliefs show a larger increase in their posterior self-beliefs—over 5 percentage points—relative to the non-pessimistic group (Table A9). However, even this group does not significantly adjust their family formation plans in response to the positive shift in beliefs (Table A10).

Second, we examine whether our experimental design—comparing *within-respondent* responses—may have muted potential treatment effects. In the main survey, respondents answered the same family formation plan questions both before and after receiving the information. This repetition could have led respondents to stick with their initial answers regardless of the information, creating an *anchoring effect* that masked treatment effects. To test this possibility, we conducted an additional survey of 400 respondents in which the family formation plan questions were asked only once, after the information was provided, and compared responses *between* treatment and control groups.¹⁹ However, even under this design, we still do not find significant treatment effects on family formation plans (Table A11).

Lastly, respondents may not interpret information about the proportion of working mothers as evidence of improved work–family compatibility. Before the main experiment, we hypothesized that learning mothers work at higher rates than expected would lead respondents to view Korean society as more supportive of work–family balance, thereby fostering positive updates in their self-beliefs and family formation plans. Yet it is also possible that respon-

¹⁹In September 2024, we administered this additional survey to 400 employees aged 19–35 through Embrain, the same survey company used for the main survey.

dents fail to make this connection. To examine this, in the additional survey we presented those who had initially underestimated population beliefs with the actual figures and asked two open-ended questions: 1) *Why did you initially estimate a lower rate of women continuing to work post-childbirth?* 2) *Korean female workers continue to work after childbirth more than you had estimated. What do you think are the reasons they actually work more than you anticipated?*²⁰

When asked why they had initially estimated a lower rate, most respondents cited the difficulty of balancing work and family in Korea. About 70% pointed to structural barriers that make it difficult for mothers to continue working, and about 16% referred to personal observations of acquaintances quitting their jobs (Figure 6). However, when presented with information showing a higher rate of working mothers, respondents did not interpret this as evidence that work–family compatibility is improving. Instead, the majority (about 80%) attributed the higher rates to financial pressures, such as high living costs and childcare expenses (Figure 7). Only 9% suggested that government policies or more equal division of household labor might be making work–family balance more feasible. In short, most respondents interpreted the higher employment rates as a reflection of financial necessity rather than enhanced work–family compatibility.

Although these results are only suggestive, they help explain why we find no treatment effects on family formation plans. Respondents appear to update their self-beliefs not because they see work–family balance as more attainable, but because they expect financial necessity will require them to remain employed. In this case, information does not translate into more positive fertility or marriage intentions. These findings suggest that simply providing accurate statistics is insufficient; efforts must also address how the information is framed and interpreted.

²⁰These questions were presented at the end of the survey.

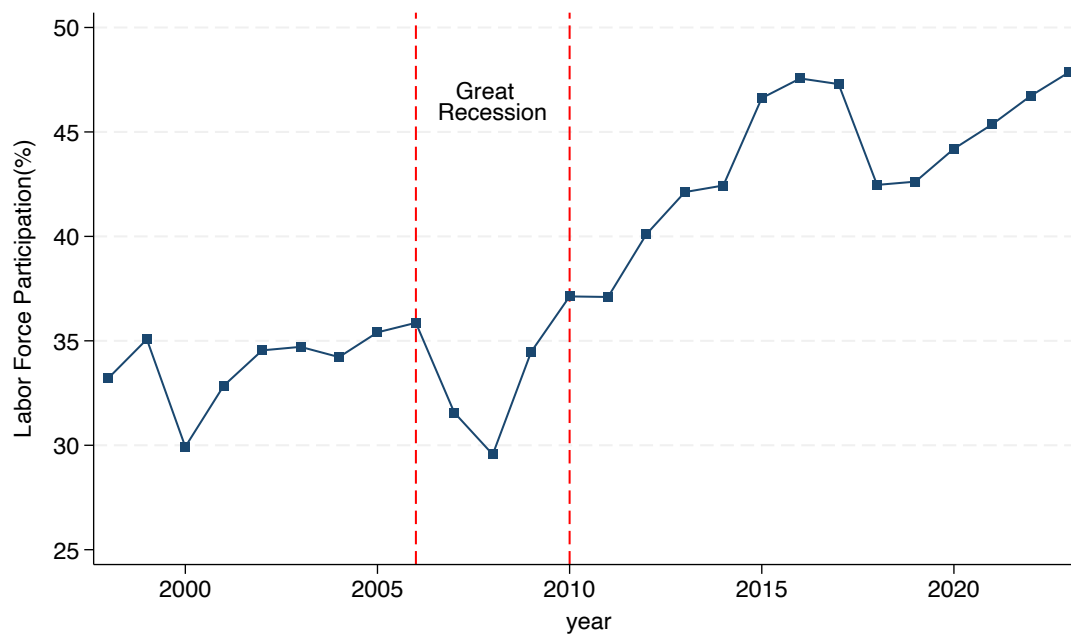
7 Conclusion

Using a representative survey from South Korea, we find that women’s expectations about their own likelihood of working after childbirth are closely linked to their fertility and marriage plans. These expectations are often pessimistic, partly because respondents believe that fewer mothers remain in the workforce after childbirth than is actually the case. Such misperceptions may contribute to Korea’s extremely low birth rate. Future research could examine whether similar patterns hold in other low-fertility countries.

Furthermore, we examine whether providing accurate information regarding women’s likelihood of working post-childbirth can change fertility and marriage plans. We find that providing objective information alone has only modest effects on plans, as respondents interpret the information as indicating that mothers are compelled to work due to financial hardships. They do not perceive the information as evidence that work-family compatibility is more achievable than previously believed. These findings shed light on the potential effectiveness of government information campaigns. Our findings suggest that simply presenting information about the situation in Korea will be insufficient to change fertility decisions. This suggests that successful campaigns require not only accurate information but also credible narratives that shape its interpretation.

Finally, our finding of widespread misperceptions about the child penalty calls for future research on the determinants of these perceptions. While it is argued that such misperceptions arise primarily because public perceptions have not kept pace with recent societal changes, the mechanisms underlying their formation remain poorly understood. Additional research is needed to explore the underlying causes of these distorted perceptions in more depth.

Figure 1: Labor Force Participation of Mothers with 0-6 years old



Notes: This figure shows the labor force participation rates of mothers with children aged 0 to 6. The data are drawn from the Korean Labor and Income Panel Study.

Figure 2: From beliefs to family formation plans

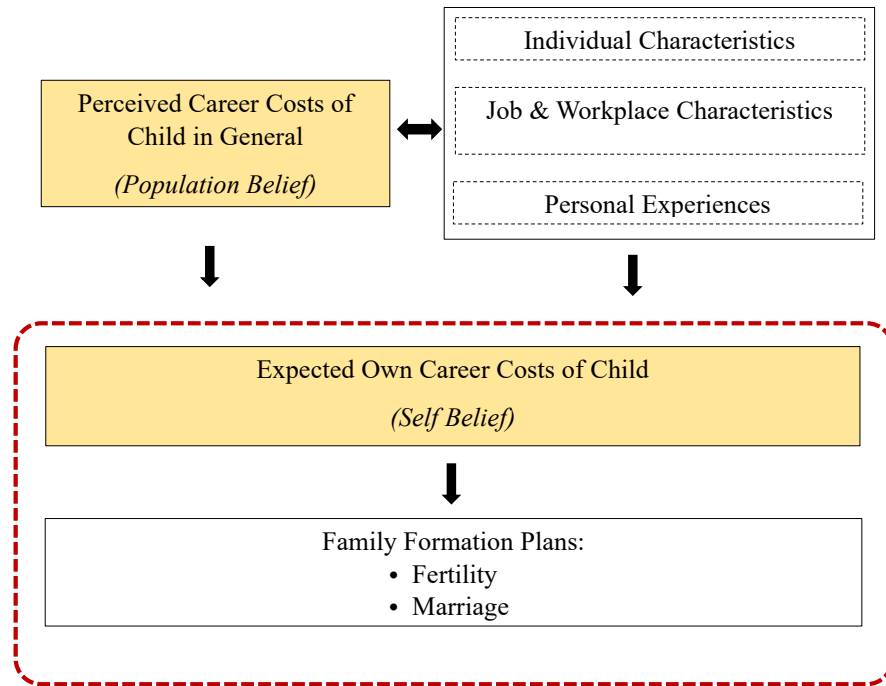
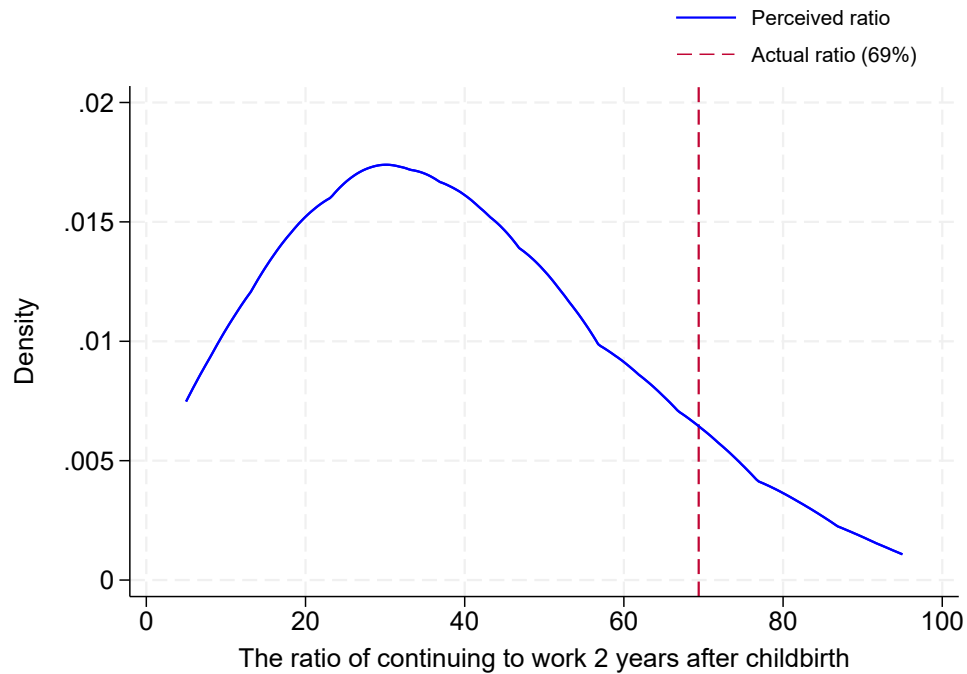
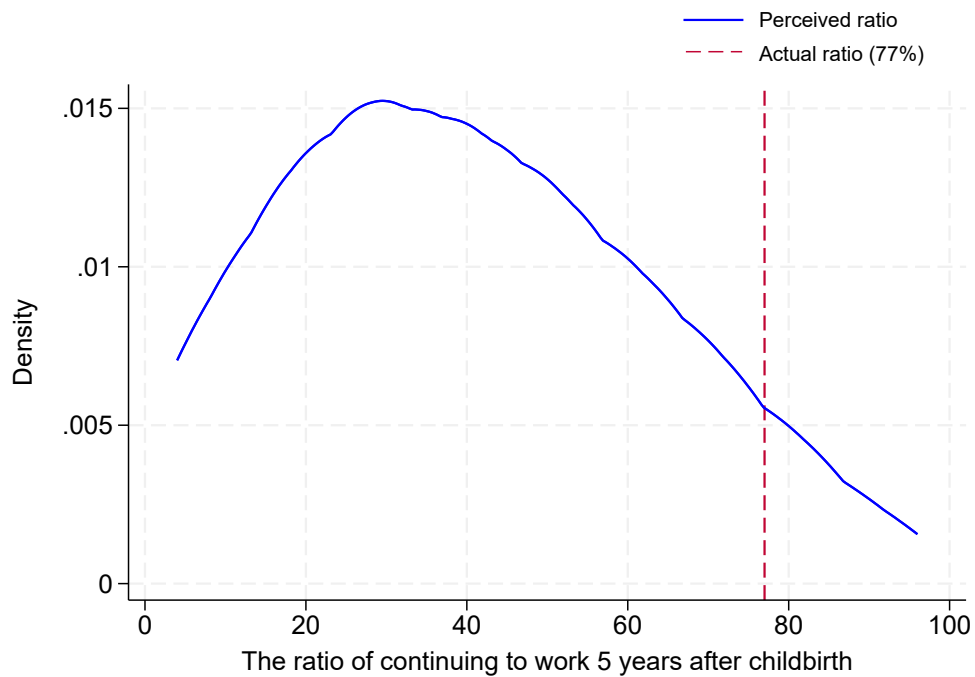


Figure 3: Perceived proportion of female workers continuing to work after childbirth

(a) After two years

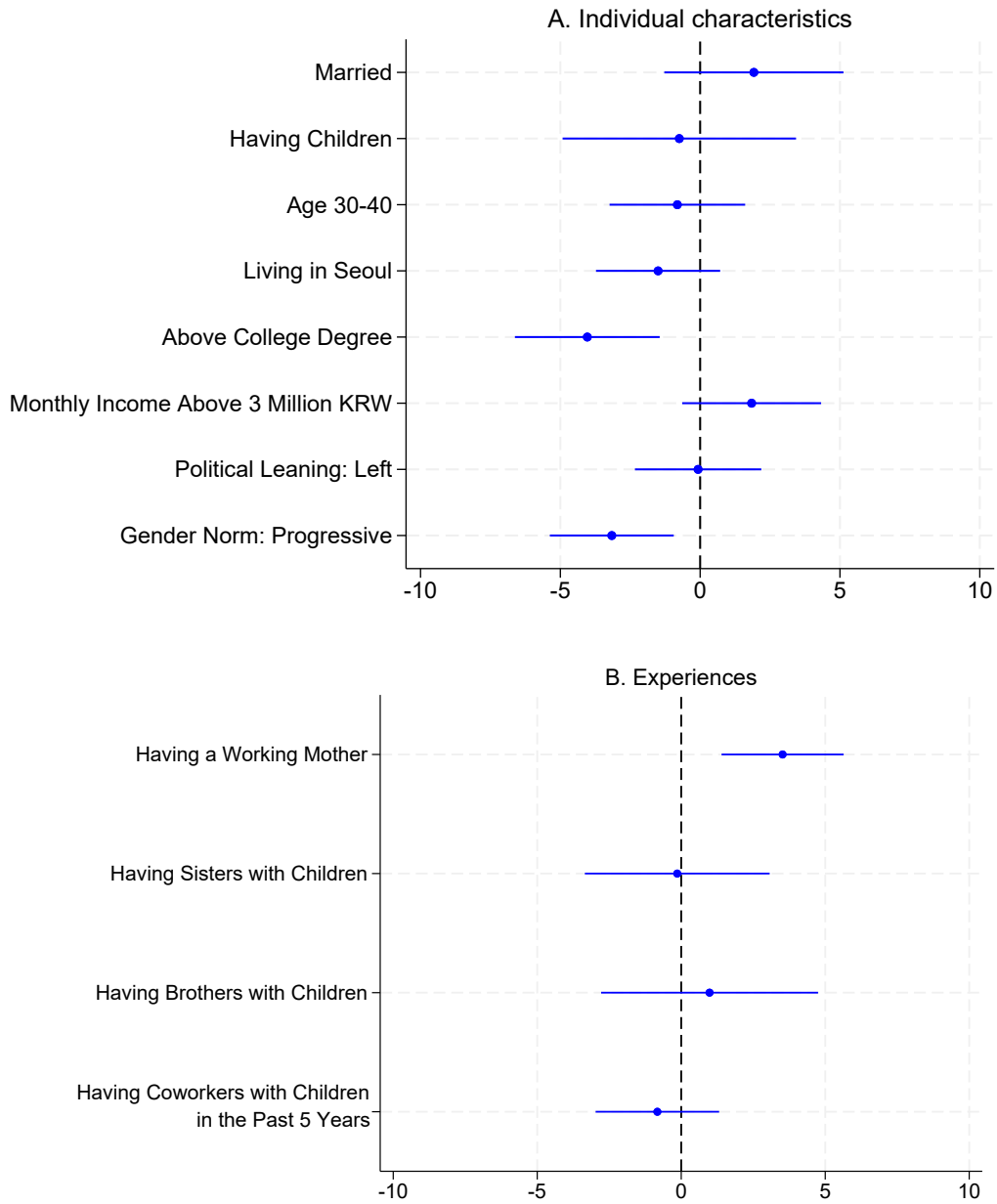


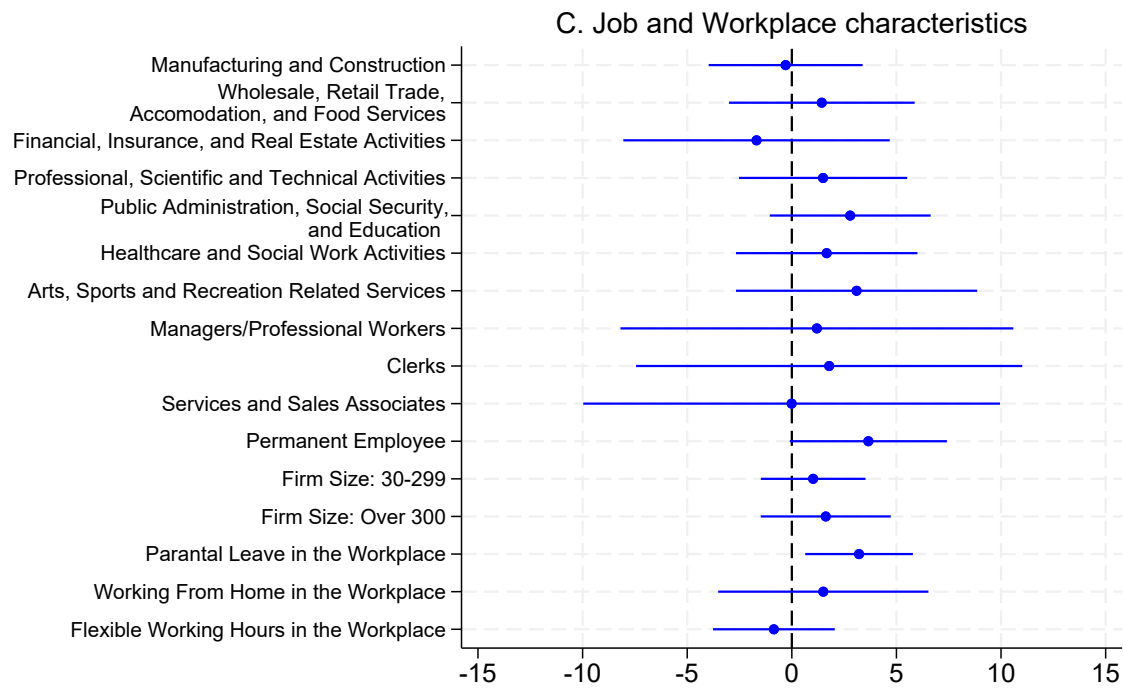
(b) After five years



Notes: The red dashed line represents the actual proportion of Korean female workers who became pregnant in 2015 and continued working after childbirth after a specified period, calculated using administrative data from Statistics Korea. The solid blue line shows the distribution of the perceived proportion for female respondents.

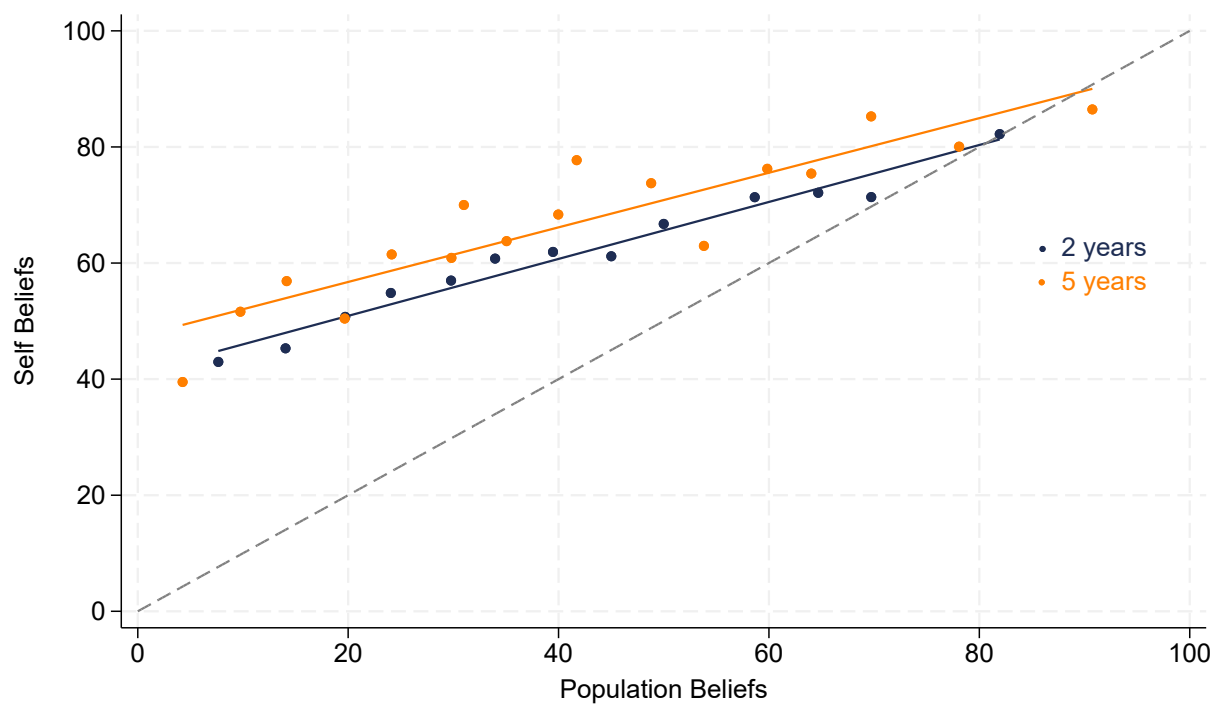
Figure 4: Correlates of population beliefs





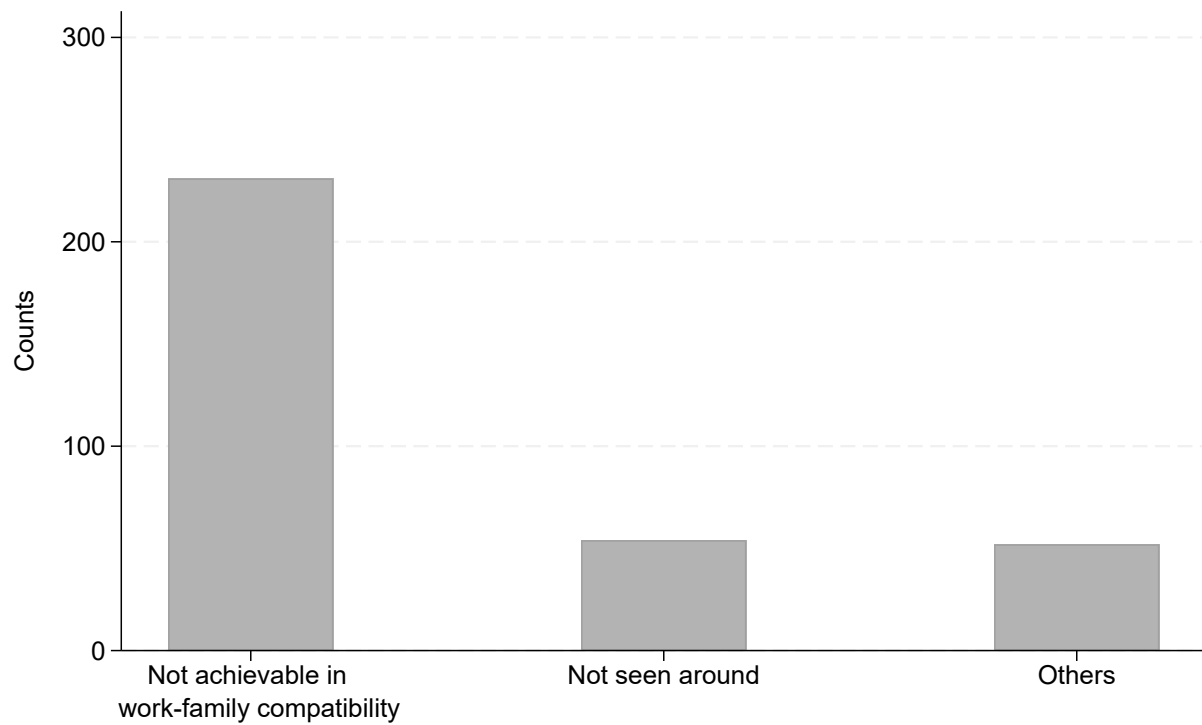
Notes: The figures show the coefficients with their associated 95% confidence interval from a regression of the population beliefs on individuals, experiences, job, and workplace characteristics indicators. popbelief 계산 2년 5년 평균. Panel A shows the results from the individual characteristics indicator. Panel B shows the results from the experiences indicator. Panel C shows the results from the job and workplace characteristics indicators. We classify an individual as having progressive gender norm if the average rating from the three questions, rated on a 5-point Likert scale, is 3.5 or higher (about half of the respondents): I disagree with the statement that a husband's job is to earn money, and a wife's job is to take care of the home and family; I disagree with the statement that when a mother works, her preschool-aged children suffer significantly; I agree with the statement that even after getting married, it is better for a woman if she continues to work.

Figure 5: Population beliefs and self beliefs



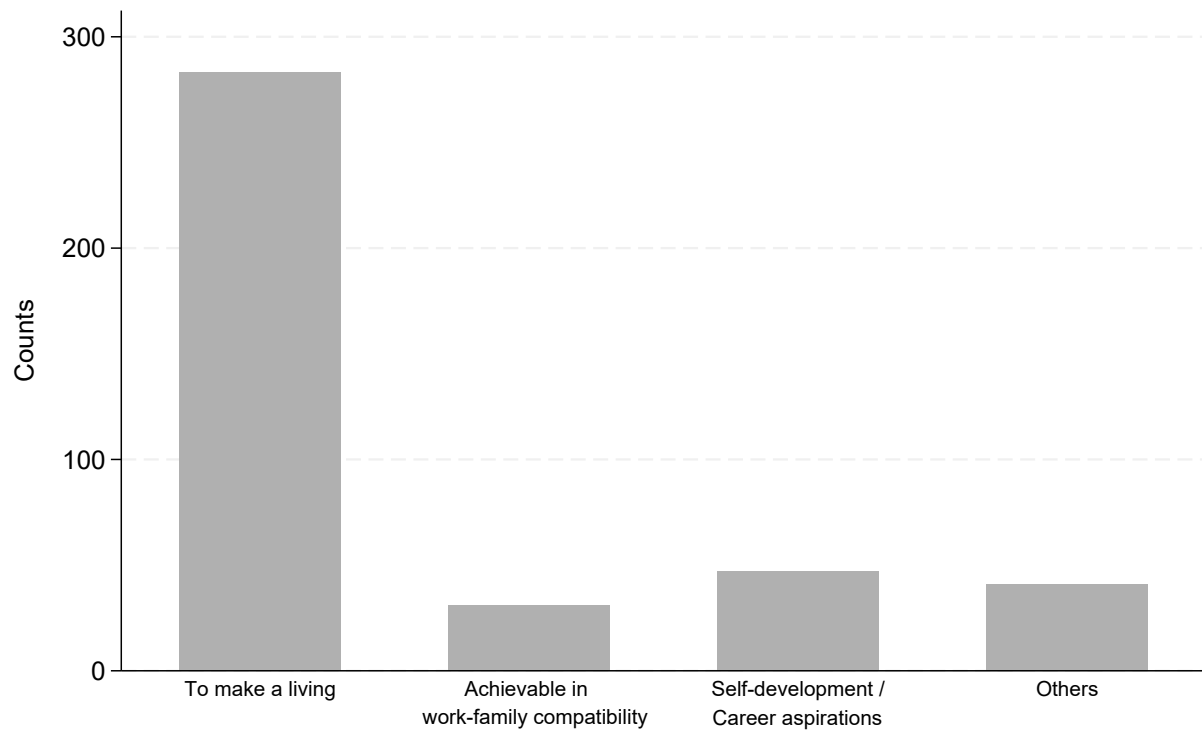
Notes: The figures display a binscatter plot of self beliefs against population beliefs for female respondents. Population beliefs represent the perceived probability of Korean female workers who continue working after childbirth. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. The two-year and five-year periods after childbirth are shown separately.

Figure 6: Perceived reasons for estimates of post-childbirth employment rates



Notes: The figure displays histograms of respondents' answers to the open-ended question: "Why did you estimate the working rate to be (reported value)%?" These questions were presented in an additional survey to 356 respondents who had initially underestimated the probability of working mothers.

Figure 7: Perceived reasons for high working rates among mothers after information



Notes: The figure displays histograms of respondents' answers to the open-ended question: "Korean female workers continue to work after childbirth more than you had estimated. What do you think are the main reasons why they actually work more than you anticipated?" These questions were presented in an additional survey to 356 respondents who had initially underestimated the probability of working mothers.

Table 1: Sample characteristics

	This Survey
Age	31.0
Living in Seoul	0.36
Above College Degree	0.73
<i>Monthly Income (10,000 KRW)</i>	304
Married	0.23
Having Children	0.10
<i>Firm Sizes:</i>	
below 10	0.21
10-99	0.44
100-299	0.15
over 300	0.20
<i>Sectors:</i>	
Manufacturing and Construction	0.23
Wholesale, Retail Trade, Accommodation and Food Service	0.09
Financial, Insurance, and Real Estate Activities	0.03
Professional, Scientific and Technical Activities	0.15
Public Administration, Social Security, and Education	0.18
Healthcare and Social Work Activities	0.13
Arts, Sports and Recreation Related Services	0.04
Other Services	0.14
Parantal leave in the workplace (%)	39
Working from home in the workplace (%)	6
Flexible working hours in the workplace (%)	17
Positive Career Prospect (%)	58
Working after marriage is better for women (%)	89
Raising children is a duty for women (%)	12
Number of respondents	1,419

Notes: The table reports descriptive statistics on the characteristics of the samples of the surveys. Column (1) shows statistics for all respondents, while Columns (2) and (3) show statistics for female and male respondents, respectively.

Table 2: Shaply Decomposition of Population Belief

Factor	
Married	4.5%
Having Children	0.4%
Age 30-40	0.5%
Living in Seoul	1.7%
College Degree	16.2%
Monthly Income	3.9%
Political Leaning	1.3%
Gender Norm	16.3%
Having a Working Mother	25.1%
Permanent Employee	4.9%
Parental Leave in the Workplace	25.3%
	100%

Notes: This table reports the Shapley decomposition of the variance in population beliefs. Each figure represents the percentage contribution of the corresponding factor to the overall explained variance.

Table 3: Anticipated career costs of children for oneself and one's spouse

Panel A. Self	
Probability of ...	
Continuing to work after 2 years	59.0 (29.2)
Continuing to work after 5 years	65.1 (29.0)
Continuing to work at the same workplace after 2 years	49.8 (33.3)
Continuing to work at the same workplace after 5 years	45.4 (34.3)
Being re-employed if they quit their job	54.2 (26.1)
Monthly wage when working after 2 years	317 (87)
Monthly wage when working after 5 years	343 (107)
Weekly labor times when working after 2 years	37.6 (9.2)
Weekly labor times when working after 5 years	44.7 (10.9)
Panel B. Spouse	
Probability of ...	
Continuing to work after 2 years	93.2 (15.3)
Continuing to work after 5 years	94.6 (14.0)
Number of individuals	1,419

Notes: The table reports descriptive statistics on the anticipated career costs of children for female and male respondents, respectively. Panel A shows expectations regarding oneself, while Panel B shows expectations regarding one's spouse. Questions other than the probability of continuing to work are measured only once after the information intervention. Therefore they are calculated using responses from the control group only. Monthly incomes are calculated by censoring values within the [5, 95] percentile range.

Table 4: Correlates of self beliefs

	(1)	(2)
	Self Belief 2 years	Self Belief 5 years
Population Belief 2 years	0.480*** (0.038)	
Population Belief 5 years		0.519*** (0.031)
Individual Characteristics:	Y	Y
Job and Workplace characteristics:	Y	Y
Experiences Indicators:	Y	Y
Other Beliefs Indicators:	Y	Y
Observations	1,401	1,401
R-squared	0.219	0.246

Notes: The table reports the coefficients from a regression of the self beliefs on population beliefs and other variables, restricted to female respondents only. Population beliefs represent the perceived probability of Korean female workers who continue working after childbirth. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Columns (1) and (2) correspond to the results about beliefs regarding the 2-year and 5-year periods after childbirth, respectively. See Table A3 for coefficients on included controls. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Family formation plans

Panel A. Fertility	
Fertility intention (0-10)	4.6 (3.7)
Probability of having a child within 3 years	28.3 (34.2)
Probability of having a child within 5 years	39.9 (36.7)
Probability of having a child within 10 years	50.0 (38.9)
Desired number of children	1.2 (0.9)
Desired timing for having children (how many years from now)	3.8 (2.6)
Number of individuals	1,420
Panel B. Marriage	
Marriage intention (0-10)	5.9 (3.5)
Probability of married within 3 years	42.5 (36.9)
Probability of married within 5 years	55.3 (36.9)
Probability of married within 10 years	64.7 (36.7)
Desired timing for marriage (how many years from now)	4.1 (3.2)
Number of individuals	1,080

Notes: The table reports descriptive statistics on the family formation plans for female and male respondents, respectively. The marriage-related question was asked only to unmarried.

Table 6: Correlates between self beliefs and family formation plans

(a) Fertility Plan

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children
Self Belief 2 years	0.008** (0.003)	0.090*** (0.031)	0.127*** (0.034)	0.133*** (0.037)	0.001 (0.001)
R-squared	0.183	0.252	0.201	0.198	0.189
Self Belief 5 years	0.012*** (0.003)	0.071** (0.030)	0.143*** (0.033)	0.182*** (0.036)	0.002*** (0.001)
R-squared	0.187	0.251	0.204	0.206	0.194
Observations	1,401	1,263	1,263	1,263	1,193
Dep.Var.Mean	4.6	28.3	39.9	50.0	1.2
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes

(b) Marriage Plan

	(1) Marriage intention (0-10)	(2) Probability of married within 3 years	(3) Probability of married within 5 years	(4) Probability of married within 10 years
Self Belief 2 years	0.015*** (0.004)	0.149*** (0.042)	0.158*** (0.042)	0.165*** (0.041)
R-squared	0.113	0.087	0.107	0.140
Self Belief 5 years	0.020*** (0.004)	0.148*** (0.040)	0.179*** (0.042)	0.213*** (0.041)
R-squared	0.124	0.087	0.111	0.151
Observations	1,062	1,062	1,062	1,062
Dep.Var.Mean	5.9	42.5	55.3	64.7
Individual Characteristics	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Correlates between self beliefs and family formation plans by career prospect

(a) Fertility Plan

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children
<i>Panel A. Positive Prospect</i>					
Self Belief 2 years	0.012** (0.005)	0.132*** (0.041)	0.164*** (0.047)	0.157*** (0.050)	0.001 (0.001)
Self Belief 5 years	0.022*** (0.005)	0.130*** (0.040)	0.213*** (0.046)	0.255*** (0.049)	0.003*** (0.001)
Observations	811	735	735	735	691
<i>Panel B. Non-positive Prospect</i>					
Self Belief 2 years	0.002 (0.005)	0.033 (0.049)	0.061 (0.051)	0.088 (0.057)	0.002 (0.002)
Self Belief 5 years	0.000 (0.005)	0.002 (0.047)	0.049 (0.049)	0.089* (0.055)	0.000 (0.002)
Observations	590	528	528	528	502

(b) Marriage Plan

	(1) Marriage intention (0-10)	(2) Probability of married within 3 years	(3) Probability of married within 5 years	(4) Probability of married within 10 years
<i>Panel A. Positive Prospect</i>				
Self Belief 2 years	0.021*** (0.005)	0.208*** (0.056)	0.201*** (0.057)	0.196*** (0.058)
Self Belief 5 years	0.032*** (0.005)	0.251*** (0.055)	0.273*** (0.056)	0.293*** (0.056)
Observations	632	632	632	632
<i>Panel B. Non-positive Prospect</i>				
Self Belief 2 years	0.007 (0.006)	0.073 (0.065)	0.090 (0.067)	0.119* (0.064)
Self Belief 5 years	0.005 (0.006)	0.029 (0.062)	0.061 (0.065)	0.119* (0.062)
Observations	430	430	430	430

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Panel A presents results for respondents who answer positively about their current career prospects, while Panel B shows results for respondents who answer neutrally or negatively about their current career prospects. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Correlates between self beliefs and family formation plans by income

(a) Fertility Plan

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children
<i>Panel A. Higher Income</i>					
Self Belief 2 years	0.016*** (0.005)	0.127** (0.041)	0.191*** (0.047)	0.248*** (0.050)	0.003** (0.001)
Self Belief 5 years	0.016*** (0.005)	0.107** (0.040)	0.205*** (0.046)	0.263*** (0.049)	0.002* (0.001)
Observations	587	514	514	514	491
<i>Panel B. Lower Income</i>					
Self Belief 2 years	0.003 (0.004)	0.065* (0.004)	0.088** (0.004)	0.057 (0.005)	0.000 (0.001)
Self Belief 5 years	0.009** (0.004)	0.056 (0.038)	0.113*** (0.042)	0.135*** (0.045)	0.001 (0.001)
Observations	814	749	749	749	702

(b) Marriage Plan

	(1) Marriage intention (0-10)	(2) Probability of married within 3 years	(3) Probability of married within 5 years	(4) Probability of married within 10 years
<i>Panel A. Higher Income</i>				
Self Belief 2 years	0.027*** (0.006)	0.188*** (0.072)	0.226*** (0.071)	0.273*** (0.069)
Self Belief 5 years	0.033*** (0.007)	0.214*** (0.073)	0.253*** (0.072)	0.312*** (0.069)
Observations	406	406	406	406
<i>Panel B. Lower Income</i>				
Self Belief 2 years	0.008* (0.005)	0.132*** (0.051)	0.126** (0.053)	0.108** (0.053)
Self Belief 5 years	0.014*** (0.005)	0.123** (0.048)	0.150*** (0.051)	0.169*** (0.051)
Observations	656	656	656	656

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Panel A presents results for respondents who earn a monthly income of 3 million KRW or above, while Panel B shows results for respondents who earn a monthly income below 3 million KRW. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Effect of treatment on beliefs

	(1) Pop Belief 2 years	(2) Pop Belief 5 years	(3) Self Belief 2 years	(4) Self Belief 5 years
<i>Panel A. Reduced form</i>				
Treated 2	18.9*** (1.1)	19.6*** (1.2)	3.6*** (1.2)	3.3** (1.4)
Treated 5	19.6*** (1.1)	21.0*** (1.2)	4.8*** (1.2)	4.4*** (1.3)
Control mean	40.2	43.2	59.3	65.3
<i>Panel B. 2SLS</i>				
Pop Belief 2 years			0.212*** (0.050)	0.195*** (0.056)
Pop Belief 5 years			0.200*** (0.050)	0.184*** (0.053)
Individual Characteristics	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes
Observations	1,336	1,336	1,336	1,336

Notes: The table reports the treatment effect of the information (2-year treatment, 5-year treatment) relative to the control group. In Panel A, Columns (1) and (2) show the treatment effects on posterior population beliefs, which indicate the perceived probability of Korean female workers who continue working after childbirth. Columns (3) and (4) show the treatment effects on female respondents' posterior self beliefs, which indicate the expected probability that respondents themselves could continue to work after childbirth. Columns (5) and (6) show the treatment effects on male respondents' posterior spouse beliefs, which indicate the expected probability that respondents' spouse could continue to work after childbirth. Panel B shows a 2SLS specification where the first stage consists of Panel A, Columns (1) and (2). Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Effect of treatment on family formation plans

(a) Fertility Plans

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children	(6) Desired timing for having children
<i>Panel A. Reduced form</i>						
Treated 2	-0.02 (0.04)	1.13 (0.84)	1.25 (0.82)	0.92 (0.78)	-0.01 (0.02)	-0.11 (0.11)
Treated 5	-0.01 (0.04)	1.51* (0.91)	0.62 (0.81)	1.41** (0.70)	-0.01 (0.02)	-0.21 (0.13)
Control mean	4.7	30.6	42.9	52.7	1.2	4.1
<i>Panel B. 2SLS</i>						
Pop Belief 2 years	-0.00 (0.00)	0.07* (0.04)	0.05 (0.04)	0.06* (0.03)	-0.00 (0.00)	-0.01 (0.01)
Pop Belief 5 years	-0.00 (0.00)	0.06* (0.04)	0.04 (0.03)	0.06* (0.03)	-0.00 (0.00)	-0.01 (0.01)
Individual Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,336	1,206	1,206	1,206	1,123	764

(b) Marriage Plans

	(1) Marriage intention (0-10)	(2) Probability of having married within 3 years	(3) Probability of having married within 5 years	(4) Probability of having married within 10 years	(5) Desired timing for marriage
<i>Panel A. Reduced form</i>					
Treated 2	-0.07 (0.08)	0.03 (0.98)	0.80 (0.97)	-0.26 (0.88)	0.01 (0.11)
Treated 5	-0.05 (0.04)	-0.07 (1.02)	-0.03 (1.03)	-0.69 (0.95)	0.25 (0.15)
Control mean	6.0	43.8	56.7	66.1	4.1
<i>Panel B. 2SLS</i>					
Pop Belief 2 years	-0.00 (0.00)	0.01 (0.04)	0.02 (0.04)	-0.00 (0.04)	0.01 (0.01)
Pop Belief 5 years	-0.00 (0.00)	0.01 (0.04)	0.02 (0.04)	-0.00 (0.04)	0.01 (0.01)
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes
Observations	1,020	1,020	1,020	1,020	651

Notes: The table reports the treatment effect of the information (2-year treatment, 5-year treatment) relative to the control group. Table (a) shows the effects of treatment on female respondents' posterior fertility plans. Table (b) shows the effects of treatment on female respondents' posterior marriage plans. For probability outcomes, sample is restricted to cases where the posterior-prior value is within the range [-90, 90]. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

References

- Adda, J., Dustmann, C., and Stevens, K. (2017). The Career Costs of Children. *Journal of Political Economy*, 125 (2): 293-337.
- Adserà, A. (2004). Changing fertility rates in developed countries. The impact of labor market institutions. *Journal of Population Economics*, 17: 17-43.
- Alesina, A., Miano, A., and Stantcheva, S. (2023). Immigration and Redistribution. *The Review of Economic Studies*, 90(1): 1-39.
- Alesina, A., Stantcheva, S., and Teso, E. (2018). Intergenerational Mobility and Preferences for Redistribution. *American Economic Review*, 108(2): 521-554.
- Baudin, T., de la Croix, D., and Gobbi, P. E. (2015). Fertility and Childlessness in the United States. *American Economic Review*, 105(6): 1852-1882.
- Bleemer, Z. and Zafar, B. (2018). Intended college attendance: Evidence from an experiment on college returns and costs. *Journal of Public Economics*, 157: 184-211.
- Boneva, T., Golin, M., Kaufmann, K., and Rauh, C. (2023). Beliefs About Maternal Labor Supply. *Working Paper*.
- Bursztyn, L., González, A. L., and Yanagizawa-Drott, D. (2020). Misperceived social norms: Women working outside the home in saudi arabia. *American economic review*, 110(10):2997–3029.
- Caucutt, E. M., Guner, N., and Knowles, J. (2002). Why Do Women Wait? Matching, Wage Inequality, and the Incentives for Fertility Delay. *Review of Economic Dynamics*, 5 (4): 815-855.
- Cho, D. and Han, J. (2024). Women’s Career Interruptions and the Declining Fertility Rate in South Korea. *KDI Focus*, 132.
- Choi, S., Shin, S., and Ro, Y. J. (2022). Work Together and Care Together?: An Evaluation of How Children Affect Parents’ Labor Supply in Korea. *Korea Labor Institute*.

- Cortes, P. and Pan, J. (2023). Children and the Remaining Gender Gaps in the Labor Market. *Journal of Economic Literature*, forthcoming.
- Costa-Ramón, A., Schaede, U., Slotwinski, M., and Brenøe, A. A. (2024). (Not) Thinking about the Future: Inattention and Maternal Labor Supply. *Working Paper*.
- Deshpande, M. and Dizon-Ross, R. (2024). The (Lack of) Anticipatory Effects of the Social Safety Net on Human Capital Investment. *American Economic Review*, 113(12): 3129-3172.
- Detting, L. J. and Kearney, M. S. (2014). House prices and birth rates: The impact of the real estate market on the decision to have a baby. *Journal of Public Economics*, 110: 82-100.
- Doepke, M., Hannusch, A., Kindermann, F., and Tertilt, M. (2023). The economics of fertility: a new era. *Handbook of Economics of the Family*, 151-254.
- Doepke, M. and Kindermann, F. (2019). Bargaining over Babies: Theory, Evidence, and Policy Implications. *American Economic Review*, 109 (9): 3264-3306.
- Duvander, A.-Z., Fahlén, S., Brandén, M., and Ohlsson-Wijk, S. (2020). Who makes the decision to have children? Couples' childbearing intentions and actual childbearing. *Advances in Life Course Research*.
- Dynan, K., Kirkegaard, J. F., and Stansbury, A. (2022). Why gender disparities persist in South Korea's labor market. *PIIE Working Paper*.
- d'Albis, H., Gobbi, P. E., and Greulich, A. (2017). Having a Second Child and Access to Childcare: Evidence from European Countries. *Journal of Demographic Economics*, 83(2): 177-210.
- Feyrer, J., Sacerdote, B., and Stern, A. D. (2008). Will the Stork Return to Europe and Japan? Understanding Fertility within Developed Nations. *Journal of Economic Perspectives*, 22(3): 3-22.
- Foerster, H., Obermeier, T., Schulz, B., and Paul, A. (2024). Job Displacement, Remarriage, and Marital Sorting. *Working Paper*.

- Fuster, A. and Zafar, B. (2023). Survey experiments on economic expectations. *Handbook of Economic Expectations*, 107-130.
- Goldin, C. (2021). Career and Family: Women’s Century-Long Journey toward Equity. *Princeton University Press*.
- Goldin, C. (2025). Babies and the Macroeconomy. *NBER Working Paper*.
- Gong, Y., Stinebrickner, R., and Stinebrickner, T. (2022). Marriage, children, and labor supply: Beliefs and outcomes. *Journal of Econometrics*, 231(1): 148-164.
- Grewenig, E., Lergetporer, P., Simon, L., Werner, K., and Woessmann, L. (2023). Can internet surveys represent the entire population? A practitioners’ analysis. *European Journal of Political Economy*, 78.
- Guner, N., Kaya, E., and Sánchez-Marcos, V. (2024). Labor Market Institutions and Fertility. *International Economic Review*, *forthcoming*.
- Haaland, I., Roth, C., and Wohlfart, J. (2023). Designing Information Provision Experiments. *Journal of Economic Literature*, 61 (1): 3-40.
- Jäger, S., Roth, C., Roussille, N., and Schoefer, B. (2024). Worker Beliefs About Outside Options. *The Quarterly Journal of Economics*, 139(3): 1505-1556.
- Kearney, M. S., Levine, P. B., and Pardue, L. (2022). The Puzzle of Falling US Birth Rates since the Great Recession. *Journal of Economic Perspectives*, 36(1): 151-176.
- Kim, A. and Hahn, Y. (2022). The motherhood effect on labour market outcomes: evidence from South Korea. *Asian-Pacific Economic Literature*, 26(2): 71-88.
- Kim, S., Tertilt, M., and Yum, M. (2024). Status Externalities in Education and Low Birth Rates in Korea. *American Economic Review*, 114(6): 1576-1611.
- Kleven, H., Landais, C., and Leite-Mariante, G. (2023). The Child Penalty Atlas. *NBER Working Paper*.

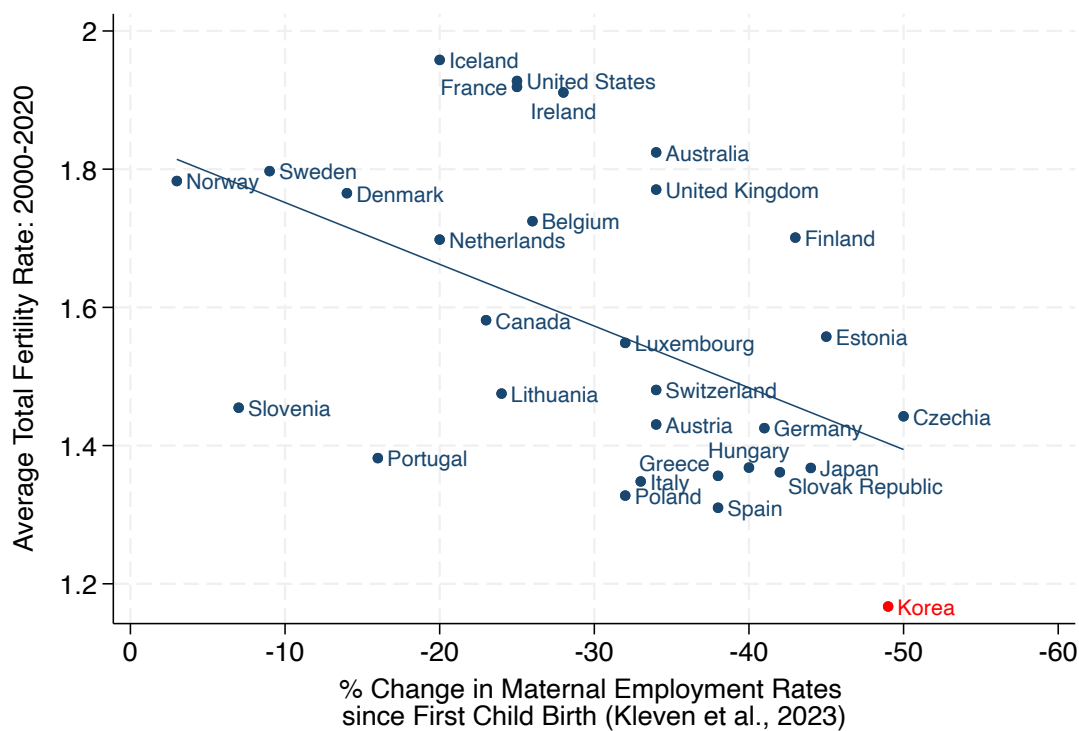
- Kleven, H., Landais, C., and Søgaaard, J. E. (2019). Children and Gender Inequality: Evidence from Denmark. *American Economic Journal: Applied Economics*, 11 (4): 181-209.
- Kuziemko, I., Pan, J., Shen, J., and Washington, E. (2018). The Mommy Effect: Do Women Anticipate the Employment Effects of Motherhood? *Working Paper*.
- Kuziemko, I., Saez, M. I. N. E., and Stantcheval, S. (2015). How Elastic Are Preferences for Redistribution? Evidence from Randomized Survey Experiments. *American Economic Review*, 105(4): 1478-1508.
- Lee, J. and Yoo, I. (2020). The Effects of Marriage and Childbearing on Labor Market Outcomes and Subjective Well-Being among Women. *Korean Journal of Labor Economics*, 43(4): 35-86 (in Korean).
- Lopes, M. (2024). Job Security and Fertility Decisions. *Working Paper*.
- Malmendier, U. and Nagel, S. (2016). Learning from Inflation Experiences. *The Quarterly Journal of Economics*, 131(1): 53-87.
- Myong, S., Park, J., and Yi, J. (2021). Social Norms and Fertility. *Journal of the European Economic Association*, 19(5): 2429-2466.
- OECD (2024a). OECD Economic Surveys: Korea 2024. *OECD Publishing, Paris*, <https://doi.org/10.1787/c243e16a-en>.
- OECD (2024b). OECD Family Database. <https://www.oecd.org/en/data/datasets/oecd-family-database.html>.
- Olivetti, C. and Petrongolo, B. (2017). The Economic Consequences of Family Policies: Lessons from a Century of Legislation in High-Income Countries. *Journal of Economic Perspectives*, 31(1): 205-230.
- Presidential Committee on Ageing Society and Population Policy (2023). <https://www.betterfuture.go.kr/front/notificationSpace/pressReleaseDetail.do?articleId=234>.

- Presidential Committee on Ageing Society and Population Policy (2024).
<https://www.betterfuture.go.kr/front/notificationSpace/pressReleaseDetail.do?articleId=338>.
- Rocha, J. M. D. and Fuster, L. (2006). Why are Fertility Rates and Female Employment Ratios Positively Correlated Across OECD Countries? *International Economic Review*, 47(4): 1187-1222.
- Rossi, P. and Xiao, Y. (2022). Spillovers in Childbearing Decisions and Fertility Transitions: Evidence from China. *Journal of the European Economic Association*, 22(1): 161-199.
- Stantcheva, S. (2023). How to Run Surveys: A guide to creating your identifying variation and revealing the invisible. *Annual Review of Economics*, 15(1): 205-234.
- Wiswall, M. and Zafar, B. (2015). Determinants of College Major Choice: Identification using an Information Experiment. *The Review of Economic Studies*, 82(2): 791–824.
- Wiswall, M. and Zafar, B. (2021). Human Capital Investments and Expectations about Career and Family. *Journal of Political Economy*, 129(5): 1361-1424.
- Yi, J. and Zhang, J. (2010). The Effect of House Price on Fertility: Evidence from Hong Kong. *Economic Inquiry*, 48(3): 635-650.

Appendices

A Additional Tables and Figures

Figure A1: Fertility and child penalty in high-income countries



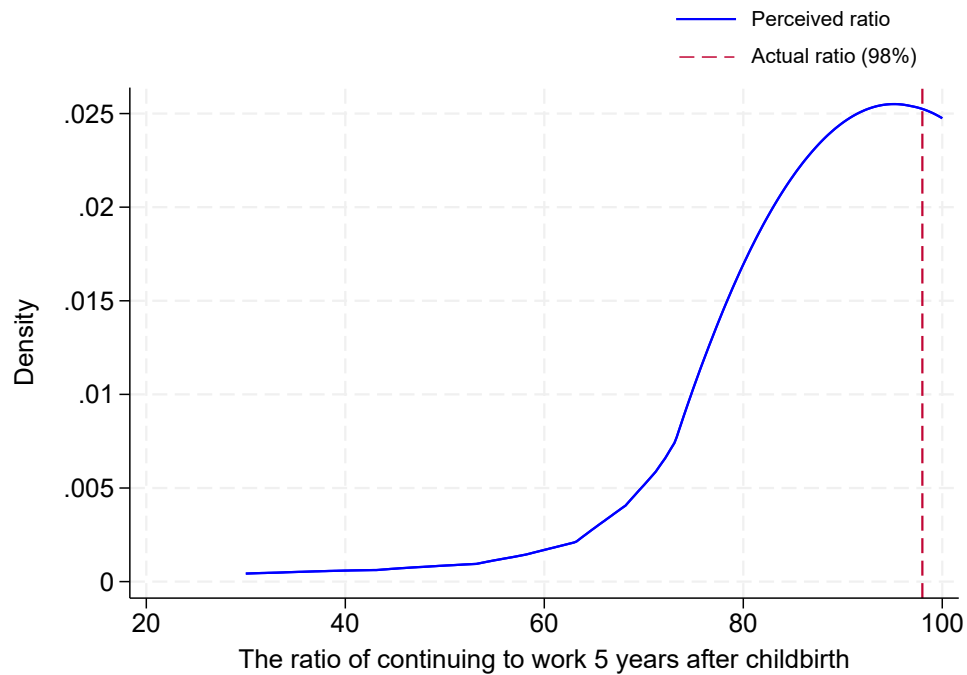
Notes: The sample consists of member countries of the OECD Development Assistance Committee (DAC), excluding New Zealand due to a lack of data on child penalty. Fertility rates for 2000–2020 are sourced from the OECD Family Database (OECD, 2024b). Child penalty in employment are sourced from the Child Penalty Atlas (Kleven et al., 2023).

Figure A2: Perceived proportion of male workers continuing to work after childbirth

(a) After two years



(b) After five years



Notes: The red dashed line represents the actual proportion of Korean male workers whose spouses became pregnant in 2015 and who continued working after childbirth after a specified period, calculated using administrative data from Statistics Korea. The solid blue line shows the distribution of the perceived proportion for female and male respondents, respectively.

Table A1: Sample Characteristics and Comparison to Other Data (Employees Aged 19-40)

	(1) This Survey	(2) KWCS 2023
Age	31.0	32.0
Living in Seoul	0.33	0.17
Above College Degree	0.72	0.56
Monthly Wage (10,000 KRW)	327	282
<i>Firm Sizes:</i>		
below 10	0.19	0.43
10-99	0.41	0.40
100-299	0.16	0.07
over 300	0.24	0.09
<i>Sectors:</i>		
Manufacturing and Construction	0.27	0.23
Wholesale and Retail Trade/Accommodation and Food Service	0.10	0.31
Financial and Insurance/Real Estate Activities	0.04	0.06
Professional, Scientific and Technical Activities	0.17	0.09
Public Administration and Social Security/Education	0.17	0.13
Healthcare/Social Work Activities	0.10	0.11
Arts, Sports and Recreation Related Services	0.04	0.02
Other Services	0.12	0.05
Married	0.22	0.44
Having Children	0.12	0.26
Number of Workers	1,419	9,988

Notes: Column (1) reports descriptive statistics on the characteristics of our survey sample, while Column (2) presents corresponding figures from the 2023 wave of the KWCS.

Table A2: Perceived proportion of female workers continuing to work after childbirth in respondent's own sector

	Mean	Median	Actual	Underestimation
<i>Manufacturing and Construction</i>				
2 years	27	20	69	94%
5 years	30	22	75	95%
<i>Wholesale, Retail Trade, Accommodation and Food Service</i>				
2 years	34	27	64	86%
5 years	37	30	73	92%
<i>Financial, Insurance, and Real Estate Activities</i>				
2 years	38	30	69	83%
5 years	45	40	77	89%
<i>Professional, Scientific and Technical Activities</i>				
2 years	44	44	72	92%
5 years	46	45	78	83%
<i>Public Administration, Social Security, and Education</i>				
2 years	59	64	81	88%
5 years	64	69	87	84%
<i>Healthcare and Social Work Activities</i>				
2 years	44	43	65	78%
5 years	47	49	75	83%
<i>Arts, Sports, and Recreation Related Services</i>				
2 years	35	28	63	87%
5 years	37	24	74	85%

Notes: This table shows how workers perceive the proportion of women in their own sector who continue working two and five years after childbirth.

Table A3: Correlates of self beliefs: Extended Table

	(1) Self Belief 2 years	(2) Self Belief 5 years
Population Belief 2 years	0.480*** (0.038)	
Population Belief 5 years		0.519*** (0.031)
Panel A: Individual Characteristics Indicators		
Married	2.201 (2.205)	1.912 (2.063)
Having Children	0.450 (3.187)	2.109 (2.974)
Age 30-40	1.535 (1.619)	0.406 (1.580)
Living in Seoul	-0.086 (1.535)	-1.671 (1.489)
Above College Degree	4.789*** (1.763)	3.305* (1.714)
Monthly Income Above 3 Million KRW	6.000*** (1.762)	3.569** (1.652)
Political Leaning: Left	-1.703 (1.555)	-1.803 (1.529)
Gender Norm: Progressive	6.567*** (1.522)	6.346*** (1.499)
Panel B: Experiences Indicators		
Having a Working Mother	3.139** (1.448)	3.196** (1.417)
Having Sisters with Children	-2.318 (2.331)	-1.185 (2.195)
Having Brothers with Children	-1.351 (2.764)	-3.340 (2.546)
Having Coworkers with Children in the Past 5 Years	1.592 (1.478)	0.733 (1.437)

(continued)

	(1) Self Belief 2 years	(2) Self Belief 5 years
Panel C: Job and Workplace characteristics Indicators		
Manufacturing and Construction	0.839 (2.483)	-1.901 (2.440)
Wholesale, Retail Trade, Accommodation and Food Service	-0.100 (3.179)	-2.088 (3.224)
Financial, Insurance, and Real Estate Activities	1.054 (4.342)	0.408 (3.789)
Professional, Scientific and Technical Activities	2.139 (2.716)	1.081 (2.675)
Public Administration, Social Security, and Education	3.274 (2.677)	5.640** (2.579)
Healthcare and Social Work Activities	2.937 (2.960)	0.252 (2.812)
Arts, Sports and Recreation Related Services	-3.675 (3.644)	-6.390* (3.796)
Managers/Professional Workers	7.115 (5.556)	6.927 (5.505)
Clerks	5.554 (5.398)	4.560 (5.402)
Services and Sales Associates	2.886 (6.139)	1.618 (6.068)
Permanent Employee	5.863** (2.637)	5.482* (2.799)
Firm Sizes: 30-299	2.260 (1.711)	0.785 (1.657)
Firm Sizes: Over 300	6.131*** (2.246)	3.404 (2.144)
Freely Available Parental Leave in the Workplace	6.115*** (1.790)	6.383*** (1.720)
Freely Available Working From Home in the Workplace	-2.912 (3.606)	-1.280 (3.367)
Freely Available Flexible Working Hours in the Workplace	3.862* (1.989)	2.071 (1.974)
Panel D: Other Beliefs Indicators		
Low Spouse Contribution to Childcare	-4.267*** (1.557)	-2.550* (1.503)
Low Grandparent Contribution to Childcare	-4.078*** (1.564)	-3.812** (1.530)
Observations	1,401	1,401
R-squared	0.219	0.246

Notes: The table reports the coefficients from a regression of the self beliefs on population beliefs and other variables, restricted to female respondents only. Population beliefs represent the perceived probability of Korean female workers who continue working after childbirth. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Columns (1) and (2) correspond to the results about beliefs regarding the 2-year and 5-year periods after childbirth, respectively. We classify an individual as having progressive gender norm if the average rating from the three questions, rated on a 5-point Likert scale, is 3.5 or higher (about half of the respondents): I disagree with the statement that a husband's job is to earn money, and a wife's job is to take care of the home and family; I disagree with the statement that when a mother works, her preschool-aged children suffer significantly; I agree with the statement that even after getting married, it is better for a woman if she continues to work. We classify low spouse contribution to childcare as respondents who believe their spouse contributes less than 50% to childcare. We classify low grandparent contribution to childcare as respondents who agree it would be difficult for grandparents to provide assistance. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Correlates of self beliefs with sector population beliefs

	(1) Self Belief 2 years	(2) Self Belief 5 years
Sector Population Belief 2 years	0.402*** (0.032)	
Sector Population Belief 5 years		0.460*** (0.027)
Panel A: Individual Characteristics Indicators		
Married	2.497 (2.323)	1.578 (2.250)
Having Children	1.207 (3.476)	3.003 (3.253)
Age 30-40	2.280 (1.762)	0.468 (1.700)
Living in Seoul	0.529 (1.611)	-1.188 (1.587)
Above College Degree	3.192* (1.924)	2.956 (1.830)
Monthly Income Above 3 Million KRW	4.231** (1.845)	2.241 (1.729)
Political Leaning: Left	-3.746** (1.674)	-3.820** (1.594)
Gender Norm: Progressive	7.179*** (1.639)	6.845*** (1.567)
Panel B: Experiences Indicators		
Having a Working Mother	4.502*** (1.555)	4.774*** (1.510)
Having Sisters with Children	-2.199 (2.676)	-0.650 (2.502)
Having Brothers with Children	-0.889 (2.915)	-3.083 (2.747)
Having Coworkers with Children	-0.293 (1.570)	-0.836 (1.498)

(continued)

	(1) Self Belief 2 years	(2) Self Belief 5 years
Panel C: Job and Workplace characteristics Indicators		
Managers/Professional Workers	7.800* (4.543)	5.950 (4.784)
Clerks	7.250* (4.374)	5.062 (4.637)
Services and Sales Associates	0.476 (5.215)	-1.471 (5.384)
Permanent Employee	10.86*** (2.724)	9.237*** (2.911)
Firm Sizes: 30-299	2.622 (1.835)	0.487 (1.777)
Firm Sizes: Over 300	7.233*** (2.338)	3.901* (2.216)
Freely Available Parental Leave in the Workplace	4.563** (1.864)	4.528** (1.805)
Freely Available Working From Home in the Workplace	-2.201 (3.384)	-0.569 (3.538)
Freely Available Flexible Working Hours in the Workplace	1.618 (2.039)	0.820 (2.028)
Panel D: Other Beliefs Indicators		
Low Spouse Contribution to Childcare	-4.370*** (1.685)	-3.593** (1.617)
Low Grandparent Contribution to Childcare	-3.853** (1.674)	-2.879* (1.621)
Observations	1,142	1,142
R-squared	0.235	0.267

Notes: The table reports the coefficients from a regression of the self beliefs on sector population beliefs and other variables, restricted to female respondents only. Sector population beliefs represent the perceived probability of Korean female workers in the respondents' sector continuing to work after childbirth. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Columns (1) and (2) correspond to the results about beliefs regarding the 2-year and 5-year periods after childbirth, respectively. We classify an individual as having progressive gender norm if the average rating from the three questions, rated on a 5-point Likert scale, is 3.5 or higher (about half of the respondents): I disagree with the statement that a husband's job is to earn money, and a wife's job is to take care of the home and family; I disagree with the statement that when a mother works, her preschool-aged children suffer significantly; I agree with the statement that even after getting married, it is better for a woman if she continues to work. We classify low spouse contribution to childcare as respondents who believe their spouse contributes less than 50% to childcare. We classify low grandparent contribution to childcare as respondents who agree it would be difficult for grandparents to provide assistance. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Other fertility and marriage outcomes

	(1)	(2)
	Desired timing for having children (year)	Desired timing for marriage (year)
Self Belief 2 years	-0.001 (0.003)	0.003 (0.006)
R-squared	0.286	0.099
Self Belief 5 years	0.002 (0.003)	0.002 (0.006)
R-squared	0.286	0.099
Observations	839	731
Dep.Var.Mean	3.8	4.1
Individual Characteristics	Yes	Yes
Job and Workplace Characteristics	Yes	Yes

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Redefine self belief as expected probability of continuing to work at same workplace

(a) Fertility Plan

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children
Self Belief 2 years: workplace	0.005 (0.005)	0.116** (0.049)	0.035 (0.053)	0.001 (0.056)	-0.002 (0.001)
R-squared	0.223	0.293	0.244	0.223	0.212
Self Belief 5 years: workplace	0.009* (0.005)	0.133*** (0.046)	0.108** (0.051)	0.102* (0.054)	-0.000 (0.001)
R-squared	0.227	0.298	0.252	0.230	0.209
Observations	469	425	425	425	393
Dep.Var.Mean	4.6	28.3	39.9	50.0	1.2
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace Characteristics	Yes	Yes	Yes	Yes	Yes

(b) Marriage Plan

	(1) Marriage intention (0-10)	(2) Probability of married within 3 years	(3) Probability of married within 5 years	(4) Probability of married within 10 years
Self Belief 2 years: workplace	0.007 (0.006)	0.183*** (0.060)	0.108* (0.062)	0.052 (0.060)
R-squared	0.154	0.153	0.157	0.192
Self Belief 5 years: workplace	0.009 (0.006)	0.179*** (0.058)	0.130** (0.059)	0.110* (0.057)
R-squared	0.157	0.153	0.162	0.199
Observations	365	365	365	365
Dep.Var.Mean	5.9	42.5	55.3	64.7
Individual Characteristics	Yes	Yes	Yes	Yes
Job and Workplace Characteristics	Yes	Yes	Yes	Yes

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could continue to work at the same workplace after childbirth. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Redefine self belief as expected probability of being re-employed

(a) Fertility Plan

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children
Self Belief: reemploy	0.019*** (0.006)	0.197*** (0.063)	0.225*** (0.070)	0.283*** (0.073)	0.000 (0.002)
Observations	469	425	425	425	393
R-squared	0.236	0.301	0.265	0.255	0.209
Dep.Var.Mean	4.6	28.3	39.9	50.0	1.2
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace Characteristics	Yes	Yes	Yes	Yes	Yes

(b) Marriage Plan

	(1) Marriage intention (0-10)	(2) Probability of married within 3 years	(3) Probability of married within 5 years	(4) Probability of married within 10 years
Self Belief: reemploy	0.027*** (0.008)	0.236*** (0.078)	0.287*** (0.080)	0.243*** (0.079)
Observations	365	365	365	365
R-squared	0.184	0.153	0.183	0.215
Dep.Var.Mean	5.9	42.5	55.3	64.7
Individual Characteristics	Yes	Yes	Yes	Yes
Job and Workplace Characteristics	Yes	Yes	Yes	Yes

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could be re-employed after childbirth if they quit their job. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Balance checks

	(1)	(2)	(3)
	2 year treat	5 year treat	Control
<i>Age:</i>			
Mean	31.0	31.1	31.0
19-24	0.05	0.05	0.04
25-29	0.45	0.45	0.46
30-34	0.24	0.23	0.25
35-40	0.26	0.27	0.25
Married	0.23	0.22	0.20
Having Children	0.13	0.11	0.12
Living in Seoul	0.32	0.33	0.33
Above College Degree	0.71	0.72	0.73
<i>Monthly Income (10,000 KRW):</i>			
Mean	333	320	329
Less than 200	0.05	0.04	0.05
200-299	0.46	0.44	0.44
300-399	0.31	0.33	0.31
More than 400	0.19	0.18	0.20
<i>Sectors:</i>			
Manufacturing and Construction	0.26	0.27	0.27
Wholesale, Retail Trade, Accommodation and Food Service	0.10	0.11	0.11
Financial, Insurance, and Real Estate Activities	0.03	0.04	0.04
Professional, Scientific and Technical Activities	0.17	0.17	0.16
Public Administration, Social Security, and Education	0.16	0.17	0.17
Healthcare and Social Work Activities	0.11	0.10	0.08
Arts, Sports and Recreation Related Services	0.05	0.03	0.04
Other Services	0.12	0.11	0.12
<i>Firm Sizes:</i>			
below 10	0.18	0.19	0.18
10-99	0.41	0.43	0.39
100-299	0.16	0.15	0.17
over 300	0.25	0.23	0.26
Number of respondents	901	911	950

Notes: The table reports descriptive statistics on the characteristics of each treatment and control group. Columns (1) and (2) show statistics for the 2-year and 5-year treatment groups, respectively, while Column (3) shows statistics for the control group.

Table A9: Heterogeneous effect of treatment on beliefs

	Self Belief 2 years	Self Belief 5 years
Treated 2	1.7 (1.3)	0.8 (1.4)
Treated 5	2.5* (1.5)	3.7*** (1.3)
Treated 2 x Pessimism	5.0** (2.4)	6.7** (2.6)
Treated 5 x Pessimism	5.3** (2.4)	2.0 (2.5)
Control mean (Non-Pessimism)	78.6	83.3
Control mean (Pessimism)	42.9	50.5
Individual Characteristics	Yes	Yes
Job and Workplace characteristics	Yes	Yes
Observations	1,336	1,336

Notes: The table reports the treatment effect of the information (2-year treatment, 5-year treatment) relative to the control group. Row 3 and 4 report the effects interacted with pessimism group. Columns (1) and (2) show the treatment effects on female respondents' posterior self beliefs, which indicate the expected probability that respondents themselves could continue to work after childbirth. For female respondents, pessimism group is defined as those with a prior two-year self belief below 65 and a five-year self belief below 75, respectively. Columns (3) and (4) show the treatment effects on male respondents' posterior spouse beliefs, which indicate the expected probability that respondents' spouse could continue to work after childbirth. For male respondents, pessimism group is defined as those with a prior two-year spouse belief below 65 and a five-year spouse belief below 75, respectively. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Heterogeneous effect of treatment on family formation plans

(a) Fertility Plans

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children	(6) Desired timing for having children
Treated 2	-0.0111 (0.05)	1.83* (0.95)	1.23 (0.86)	0.10 (0.84)	-0.04* (0.02)	-0.16 (0.12)
Treated 5	0.02 (0.05)	1.77 (1.14)	0.69 (0.83)	0.34 (0.65)	-0.01 (0.02)	-0.09 (0.11)
Treated 2 x Pessimism	-0.03 (0.07)	-1.63 (1.12)	-0.01 (1.22)	1.25 (1.16)	0.09*** (0.03)	0.10 (0.12)
Treated 5 x Pessimism	-0.07 (0.07)	-1.11 (1.18)	-0.94 (1.05)	1.39* (0.81)	0.00 (0.03)	-0.26 (0.26)
Control mean (Non-Pessimism)	4.8	32.1	44.7	55.1	1.2	4.2
Control mean (Pessimism)	4.5	29.2	41.1	50.3	1.2	3.9
Individual Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,336	1,206	1,206	1,206	1,123	764

(b) Marriage Plans

	(1) Marriage intention (0-10)	(2) Probability of having married within 3 years	(3) Probability of having married within 5 years	(4) Probability of having married within 10 years	(5) Desired timing for marriage
Treated 2	-0.07 (0.10)	0.16 (1.25)	0.45 (1.23)	-1.26 (1.14)	-0.02 (0.14)
Treated 5	0.11 (0.09)	0.47 (1.21)	0.28 (1.25)	-1.08 (1.17)	-0.00 (0.15)
Treated 2 x Pessimism	0.01 (0.11)	-0.26 (1.43)	0.77 (1.49)	2.15 (1.34)	0.05 (0.18)
Treated 5 x Pessimism	-0.35*** (0.12)	-1.41 (1.66)	-1.35 (1.78)	0.83 (1.63)	0.53** (0.26)
Control mean (Non-Pessimism)	6.3	41.7	54.7	63.8	4.7
Control mean (Pessimism)	5.6	39.4	51.6	62.7	4.1
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes
Observations	1,020	1,020	1,020	1,020	1,020

Notes: The table reports the treatment effect of the information (2-year treatment, 5-year treatment) relative to the control group. Table (a) shows the effects of treatment on female respondents' posterior fertility plans. Table (b) shows the effects of treatment on female respondents' posterior marriage plans. Row 3 and 4 report the effects interacted with pessimism group. pessimism group is defined as those with a prior two-year self belief below 65 and a five-year self belief below 75, respectively. For probability outcomes, sample is restricted to cases where the posterior-prior value is within the range [-90, 90]. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Additional treatment effect on family formation plans

(a) Fertility Plans

	(1) Fertility intention (1-5)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children	(6) Desired timing for having children
Treated	-0.03 (0.16)	0.04 (4.05)	0.43 (4.46)	1.12 (4.57)	-0.03 (0.09)	0.16 (0.27)
Control mean	3.2	28.5	40.0	49.3	1.2	3.7
Individual Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Observations	298	298	298	298	229	229

(b) Marriage Plans

	(1) Marriage intention (1-5)	(2) Probability of having married within 3 years	(3) Probability of having married within 5 years	(4) Probability of having married within 10 years	(5) Desired timing for marriage
Treated	0.03 (0.17)	3.47 (4.82)	1.95 (4.54)	2.38 (4.58)	0.60 (0.71)
Control mean	2.6	39.4	52.8	62.8	3.7
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes
Observations	261	261	261	261	230

Notes: The table reports the treatment effect of the information relative to the control group in additional survey. Table (a) shows the effects of treatment on female respondents' posterior fertility plans. Table (b) shows the effects of treatment on female respondents' posterior marriage plans. For probability outcomes, sample is restricted to cases where the posterior-prior value is within the range $[-90, 90]$. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2 Simple Model of Fertility Choice

To understand how the expectations about career cost affect the current period's family formation decisions among working-age individuals, we consider a simple two-stage model of fertility choice based on [Doepke et al. \(2023\)](#). For simplicity, we focus solely on the fertility choice and abstract from period specific occupational choice, as modeled in other structural papers such as [Adda et al. \(2017\)](#). In our model, couple's who has already chosen their occupation and education level enters a stage in which they first decides whether to have a child (fertility choice, b) in Period 1, and then choose their consumption in Period 2 conditional on the realized birth outcome from Period 1. The couple's maximization problem in Period 2, conditional on realized birth outcome b is:

$$\begin{aligned} V(b) = \max \log(c) + \delta b \quad \text{s.t.} \quad & c + \theta b = (1 - \phi_f b) \cdot w_f + (1 - \phi_m b) \cdot w_m \\ & \text{s.t.} \quad l_f + \phi_f b = 1, \quad l_m + \phi_m b = 1 \end{aligned}$$

Their utility in Period 2 depends on consumption (c) and the existence of a child (b). δ represents the weight attached to the utility derived from children. They maximize this utility subject to time and budget constraints. With a child, they face time constraints on how much time they can spend on working: $l_f + \phi_f b = 1$ and $l_m + \phi_m b = 1$, for the female and male partner, respectively. They have to spend ϕ amount of time outside the labor market on childcare if they have a child, and make earning by providing l unit of labor. They earn a wage w , which is assumed to decrease with the amount of time spent outside the labor market, $\frac{\partial w}{\partial \phi} < 0$. This assumption captures the permanent impact of childbearing on earning trajectory as illustrated in child penalty literature ([Adda et al., 2017](#)). θb is a monetary cost related to childcare (e.g. education and other necessities). In this model, ϕ captures the magnitude of overall career disruption due to childbearing. It's easy to see that utility declines with greater career disruption ($\frac{\partial V(b)}{\partial \phi} < 0$), as both labor supply, l , and wage level, w , are decreasing in ϕ .

In Period 1, the couple's choose to have a child ($b = 1$) if and only if they expected utility from having a child exceeds that of remaining childless:

$$E[V(1; \phi)|\Omega] \geq E[V(0)|\Omega],$$

where Ω represent an information set that includes one's perception of child-rearing environment. We assume that Ω represents the *subjective* information set. People may have pessimistic or optimistic views on how favorable environment is for work-family compatibility. If an individuals hold a pessimistic view (Ω_p), then their expected level of ϕ will be higher than true ϕ^* , which lowers the expected utility from having a child ($E[V(1; \phi)|\Omega_p] < E[V(1; \phi^*)]$).

Consequently, pessimistic beliefs about work-family compatibility may lead to an optimal choice of childlessness ($b = 0$) in Period 1, even in cases where the optimal choice under accurate beliefs (ϕ^*) would have been to have a child ($b = 1$).

3 Re-weighted Sample

As a robustness check, we compute weights to match our sample to the KWCS in Table A1. We divide the sample into 32 cells based on the following characteristics: age (2, age 19-29 vs. age 30-40) $\times$ residential area (2, Seoul vs. others) $\times$ education level (2, with vs. without a college degree) $\times$ income (2, below vs. above the median income (2.82 million KWR)) $\times$ firm size (2, fewer than 10 vs. 10 or more employees). For each cell i , we computed a weight w_i as follows:

$$w_i = \frac{P(X = x_i)}{S(X = x_i)}$$

where $P(X = x_i)$ denotes the share of cell i in the overall population (based on the KWCS), and $S(X = x_i)$ denotes the share of cell i in our survey sample.

In this section, we report the correlation and treatment effect estimates using the re-weighted sample. Re-weighting does not significantly affect the main results.

Table C1: Correlates between self beliefs and family formation plans: Re-Weighted Sample

(a) Fertility Plan

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children
Self Belief 2 years	0.009** (0.004)	0.202*** (0.045)	0.146*** (0.052)	0.121** (0.056)	0.000 (0.001)
R-squared	0.140	0.206	0.157	0.119	0.245
Self Belief 5 years	0.008** (0.004)	0.152*** (0.047)	0.164*** (0.050)	0.161*** (0.053)	0.000 (0.001)
R-squared	0.138	0.196	0.162	0.126	0.245
Observations	1,401	1,263	1,263	1,263	1,193
Dep.Var.Mean	4.6	28.3	39.9	50.0	1.2
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes

(b) Marriage Plan

	(1) Marriage intention (0-10)	(2) Probability of married within 3 years	(3) Probability of married within 5 years	(4) Probability of married within 10 years
Self Belief 2 years	0.007 (0.005)	0.155*** (0.055)	0.141** (0.055)	0.131** (0.056)
R-squared	0.112	0.164	0.155	0.135
Self Belief 5 years	0.008* (0.005)	0.138*** (0.053)	0.170*** (0.050)	0.162*** (0.051)
R-squared	0.112	0.162	0.162	0.142
Observations	1,062	1,062	1,062	1,062
Dep.Var.Mean	5.9	42.5	55.3	64.7
Individual Characteristics	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes

Notes: The table reports the coefficients from a regression of the family formation plans on self beliefs and other control variables, restricted to female respondents only. Self beliefs represent the expected probability that respondents themselves could continue to work after childbirth. Self beliefs regarding the 2-year and 5-year periods after childbirth are calculated, respectively. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: Effect of treatment on beliefs: Re-Weighted Sample

	Pop Belief 2 years	Pop Belief 5 years	Self Belief 2 years	Self Belief 5 years
<i>Panel A. Reduced form</i>				
Treated 2	15.9*** (1.6)	17.2*** (1.8)	2.7 (1.7)	0.8 (1.9)
Treated 5	17.8*** (1.7)	19.7*** (1.8)	6.5*** (1.6)	5.4*** (1.8)
Treated 2 x Men	1.3 (2.1)	1.7 (2.2)		
Treated 5 x Men	0.9 (2.1)	3.9* (2.2)		
Control mean	40.2	43.2	59.3	65.3
<i>Panel B. 2SLS</i>				
Pop Belief 2 years			0.253*** (0.078)	0.170* (0.087)
Pop Belief 5 years			0.150 (0.134)	0.191 (0.135)
Individual Characteristics	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes
Observations	1,336	1,336	1,336	1,336

Notes: The table reports the treatment effect of the information (2-year treatment, 5-year treatment) relative to the control group. In Panel A, Columns (1) and (2) show the treatment effects on posterior population beliefs, which indicate the perceived probability of Korean female workers who continue working after childbirth. Columns (3) and (4) show the treatment effects on female respondents' posterior self beliefs, which indicate the expected probability that respondents themselves could continue to work after childbirth. Columns (5) and (6) show the treatment effects on male respondents' posterior spouse beliefs, which indicate the expected probability that respondents' spouse could continue to work after childbirth. Panel B shows a 2SLS specification where the first stage consists of Panel A, Columns (1) and (2). Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Effect of treatment on family formation plans: Re-Weighted Sample

(a) Fertility Plans

	(1) Fertility intention (0-10)	(2) Probability of having a child within 3 years	(3) Probability of having a child within 5 years	(4) Probability of having a child within 10 years	(5) Desired number of children	(6) Desired timing for having children
Treated 2	-0.06 (0.10)	1.14 (1.21)	0.77 (1.32)	0.55 (1.21)	-0.02 (0.02)	0.05 (0.15)
Treated 5	-0.01 (0.06)	0.96 (1.30)	0.90 (1.12)	1.69 (1.08)	-0.03 (0.02)	-0.22 (0.22)
Control mean	4.7	30.6	42.9	52.7	1.2	4.1
Individual Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,336	1,206	1,206	1,206	1,123	764

(b) Marriage Plans

	(1) Marriage intention (0-10)	(2) Probability of having married within 3 years	(3) Probability of having married within 5 years	(4) Probability of having married within 10 years	(5) Desired timing for marriage
Treated 2	-0.13 (0.13)	-1.90 (1.18)	0.31 (1.29)	-0.13 (1.06)	0.05 (0.11)
Treated 5	-0.26* (0.15)	-0.65 (1.23)	-2.07 (1.66)	-0.91 (1.35)	0.18 (0.14)
Control mean	4.7	30.6	42.9	52.7	1.2
Individual Characteristics	Yes	Yes	Yes	Yes	Yes
Job and Workplace characteristics	Yes	Yes	Yes	Yes	Yes
Observations	1,020	1,020	1,020	1,020	651

Notes: The table reports the treatment effect of the information (2-year treatment, 5-year treatment) relative to the control group. Table (a) shows the effects of treatment on female respondents' posterior fertility plans. Table (b) shows the effects of treatment on female respondents' posterior marriage plans. For probability outcomes, sample is restricted to cases where the posterior-prior value is within the range [-90, 90]. Individual characteristics control includes a respondent's age, living in Seoul, having children, marital status, education, monthly income, political leaning, and gender norm. Job and Workplace characteristics control includes a respondent's sector, occupation, employment status, and firm size. Heteroskedacity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Questionnaire

4.1 Korean Version

The original survey questionnaire in Korean is available in the following link:

[Questionnaire-Korean Version](#)

4.2 English Version

The translated survey questionnaire in English is available in the following link:

[Questionnaire-English Version](#)